

SUPPLEMENT TO “NEURAL BELLMAN OPERATORS”

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This supplement supplies proofs and implementation details for the R48 revision, retaining the R44–R47 results: centered economic transport, direct full-policy certification, retained-policy recertification, and coupled policy-cost comparisons. It reports all new comparison intervals and separates mathematical, numerical, and statistical guarantees. The constructive neural backend has its own complete primitive-budget proof below. Complete earlier proofs and economic applications are materialized unchanged in the current source tree.

1. NOTATION AND ECONOMIC DOMAINS

Dates are $0, \dots, T$; $B_0 = 1$ and $B_t = \prod_{j < t} \beta_j$ with $\beta_j > 0$. The state domain is invariant under all admissible actions and the declared innovation law. The fixed-policy operator is $\mathcal{T}_t^\pi v(x) = c_t(x, \pi_t(x)) + \beta_t \rho_t(v(F_t(x, \pi_t(x), Z)))$. The Bellman operator is its infimum over feasible current actions. Every cash-invariance and monotonicity statement is restricted to a common domain containing the values, fitted functions, and constant translates used in the comparison. This requirement is substantive for recursive utility.

For compact continuous expected-cost problems with continuous feasible correspondences, continuous integrals, and measurable minimizers, backward induction produces the Bellman optimum. An adapted policy cannot lower its conditional continuation cost below the Bellman value at any date. Induction therefore proves optimality against history-dependent admissible controls, rather than only Markov grid policies. The same reasoning applies to the stipulated monotone recursive evaluation when its conditional recursion and measurable selection conditions hold. The Gaussian entropic construction in the original article uses separate moment margins rather than compactness.

2. CENTERED DECISION AND TRANSPORT PROOFS

Write $H(a) = r_\theta(s, a) + S_\theta(T_\theta(s, a))$ and $\hat{H}(a) = r_\theta(s, a) + \hat{S}_0(T_\theta(s, a))$. Then $\hat{H} - H = e \circ T_\theta - \Delta_\theta \circ T_\theta$. For an arbitrarily near-optimal true action a^* ,

$$H(a^*) - H(\hat{a}) \leq \delta + (H - \hat{H})(a^*) - (H - \hat{H})(\hat{a}).$$

Bounding the two oscillations and letting the approximation to a^* vanish proves the centered bound, even if a supremum is not attained. For the finite menu let $\bar{e} = \sum_j p_j e_j$. For distinct i, j , weighted Cauchy–Schwarz gives

$$|e_i - e_j|^2 = \left| \sum_k p_k (e_k - \bar{e}) \left(\frac{\mathbf{1}_{k=i}}{p_i} - \frac{\mathbf{1}_{k=j}}{p_j} \right) \right|^2 \leq R_p (p_i^{-1} + p_j^{-1}).$$

Maximizing over i, j and bounding the continuation perturbation by $2b_\theta$ proves the menu statement. It is a statement about a common menu; a method-specific pair of actions cannot substitute for the full set.

For transport, couple the future economies with the same innovations, starting from the same postdecision state. The assumed transition bounds imply pathwise $\|X_{k+1}^\theta - X_{k+1}^0\| \leq L_k \|X_k^\theta - X_k^0\| + \epsilon_{\theta,k}$. Thus $\|X_k^\theta - X_k^0\| \leq d_k$ by induction. Each future reward differs by at most $\eta_{\theta,k} + H_k d_k$. Summation with nonnegative weights and expectation prove the transport bound. If the recursion is nonlinear but sup-norm Lipschitz on a common utility domain, add and subtract the anchor map at the changed continuation to obtain $b_k \leq v_k + \beta_k b_{k+1}$; unwind this inequality backward.

These proofs depend only on the declared economic perturbation class. They do not assert that all policy changes, altered constraints, or opponent deviations preserve the same domain or constants.

3. SIGNED FULL-POLICY CERTIFICATE AND RETAINED ACTORS

Suppose $\ell_t \leq f_t - \mathcal{T}_t f_{t+1} \leq u_t$ and $\mathcal{T}_t^\pi f_{t+1} - \mathcal{T}_t f_{t+1} \leq \eta_t$ for $t < T$, with terminal enclosure $\ell_T \leq f_T - g \leq u_T$. For each date let $B_{t,k} = \prod_{j=t}^{k-1} \beta_j$ and set

$$a_t = \sum_{k=t}^{T-1} B_{t,k} u_k + B_{t,T} u_T, \quad b_t = \sum_{k=t}^{T-1} B_{t,k} (\eta_k - \ell_k) - B_{t,T} \ell_T.$$

We prove $V_t^* \geq f_t - a_t$ and $J_t^\pi \leq f_t + b_t$ simultaneously. The terminal claim is the terminal enclosure. For the lower induction,

$$V_t^* = \mathcal{T}_t V_{t+1}^* \geq \mathcal{T}_t(f_{t+1} - a_{t+1}) \geq f_t - u_t - \beta_t a_{t+1} = f_t - a_t.$$

For the upper induction,

$$J_t^\pi = \mathcal{T}_t^\pi J_{t+1}^\pi \leq \mathcal{T}_t^\pi(f_{t+1} + b_{t+1}) \leq f_t - \ell_t + \eta_t + \beta_t b_{t+1} = f_t + b_t.$$

Subtraction yields the theorem. Additive shifts d_t change each residual by $d_t - \beta_t d_{t+1}$ and the terminal residual by d_T ; widths and action gaps are unchanged. This proves the invariance without assuming a contraction factor less than one.

For retained-policy recertification, new residual enclosures are substituted in exactly the same induction. The old actor allowance remains valid because its hypothesis is a uniform statement about the original actor and the unchanged functions. It has no dependence on the mesh on which the new residual happens to be validated. The terminal allowance can also be retained. Intersecting two valid residual intervals preserves validity and cannot increase width. In the executed diagnostics the new residuals are used directly, with the old actor and terminal enclosures; the record contains every component needed to reconstruct the bound.

The zero-shift case does not imply that constant residuals disappear from all value brackets. They disappear from the policy-loss bound through upper-minus-lower cancellation. Keeping signed endpoints is consequently important for direct cost scores as well as for sharp policy bounds.

4. VERIFICATION GEOMETRY AND NUMERICAL ENCLOSURES

4.1. *The scalar primitive economy*

The scalar state is $x \in [0, 1]$, action $a \in [0, 1/4]$, horizon four, and discount $15/16$. The transition is

$$F(x, a, z) = \text{clip}_{[0,1]} \{ (13/16)x + a + (1/16)x(1 - x) + z \},$$

with $z \in \{-1/16, 0, 1/16\}$ and probabilities $(1/4, 1/2, 1/4)$. The cost is

$$c(x, a) = (x - 11/16)^2 + pa^2 + 4a^4 + 2(3/8 - x)_+^2,$$

and terminal cost is $g(x) = 2(x - 11/16)^2 + 2(3/8 - x)_+^2$. The two evaluations are expectation and the entropic certainty equivalent $\rho_\theta(v) = \theta^{-1} \log \mathbb{E} e^{\theta v}$ with $\theta = 1$; expectation is $\theta = 0$. The discrete law here is an economic primitive, unlike the continuous uniform law of the coupled experiment.

For a scalar ReLU network, exact dyadic coefficients permit rational breakpoints and slopes. The largest absolute slope bounds its Lipschitz constant L_t . The Bellman state and action constants used by the verifier are

$$L_t^x = 2.875 + (15/16)(0.875)L_{t+1}, \quad L_t^a = p/2 + 1/4 + (15/16)L_{t+1}.$$

For $N + 1$ state nodes and $A + 1$ action nodes, the action cover allowance is $L_t^a / (8A)$ and the residual state allowance is $(L_t + L_t^x) / (2N)$. Each residual endpoint includes the state allowance; the nodal Bellman lower bound includes the action cover. The nearest-node actor allowance adds the selected nodal action gap, action cover, and L_t^x / N . All operations are outward enclosed.

The terminal residual is piecewise quadratic. Its extrema occur at exact network breakpoints, the cost kink, endpoints, or stationary points in each piece. The implementation enumerates these candidates using rational arithmetic and rounds the extrema outward. It therefore does not replace an exact terminal calculation with an uncharged sampled check.

4.2. The coupled action and state certificates

For fixed x , the coupled action objective $q(a)$ is μ -strongly convex on the simplex, with $\mu = 2p - 1/2$. For any feasible proposal a and nonnegative multiplier ν , the global quadratic lower model yields

$$\inf_{b \geq 0, \mathbf{1}'b \leq h} q(b) \geq q(a) - \nabla q(a)'a + \frac{\mu}{2} \|a\|^2 - \nu h - \frac{1}{2\mu} \sum_j (\min\{\nabla_j q(a) - \mu a_j + \nu, 0\})^2, \quad h = 1/8.$$

Indeed, strong convexity gives a quadratic lower model; adding $\nu(\mathbf{1}'b - h)$ gives a lower bound on the constrained infimum, and minimizing the separable quadratic over $b \geq 0$ gives the displayed expression. Any nonnegative multiplier is valid.

Interval lower evaluation and the feasible proposal's upper value enclose the continuous-action infimum without trusting numerical first-order convergence.

Let δ bound the nodal primal–dual gap, and let L_{xa} bound the change of the action gradient with state. A node action with gap δ is within $\sqrt{2\delta/\mu}$ of the node optimizer. At a state distance r , the gradient perturbation is at most $L_{xa}r$. The loss of retaining the node action is bounded by

$$\eta = \delta + (L_{xa}r)\sqrt{2\delta/\mu} + \frac{(L_{xa}r)^2}{2\mu}.$$

This follows by combining strong convexity at the node with the cross-state change of objective differences and maximizing the resulting linear-minus-quadratic expression in the distance to the node optimizer. On the square tensor grid, $r = \sqrt{2}/(2N)$. Rounded acquisition adds its own state-distance allowance before applying the formula. The ideal policy and rounded-state policy bounds are separately named in the original records.

If the gradient of the fitted current continuation has Lipschitz constant M_t and the Bellman envelope has gradient Lipschitz constant H_{t+1} , bilinear interpolation on cells of side $1/N$ has residual error at most $(M_t + H_{t+1})/(4N^2)$. To see this, condition successively on a coordinate and use the one-dimensional linear-interpolation error $Lh^2/8$ for a function with Lipschitz derivative. Summing two coordinate errors gives $Lh^2/4$. The source-bound derivative envelopes, rather than empirical finite differences, supply these constants. The terminal allowance uses $(M_T + 18)/(4N^2)$.

In the retained-policy diagnostic, these residual terms are refined at $N = 256$, but the actor term remains the one certified for $N = 128$. The record may also contain the fine-grid proposed actor's allowance as part of the inherited verifier output. That allowance is not used to claim an unchanged-policy result.

4.3. Arithmetic assumptions and implementation boundary

The inherited interval implementation applies outward rounding to elementary binary operations and encloses its exponential and logarithm approximations by explicit series remainders. Stored network and actor coefficients are interpreted as exact dyadics. Analytic innovation moments are enclosed using interval arithmetic,

including cancellation. Exact rational breakpoint calculations and independently checked square roots are used where specified.

The resulting certificates concern the defined policies, with numerical enclosures around their evaluation. The original rounded-state and execution accounts remain separate. The new retained-policy bound does not silently assert that an arbitrary hardware implementation with different state acquisition, optimizer, or clipping conventions inherits the same target. A deployed implementation must satisfy the execution assumptions or add its own allowance.

5. PAIRED POLICY COSTS: PROBABILITY AND NUMERICAL ERROR

5.1. *An unbiased score with certificate-derived range*

Under expectation and a fixed policy, the cost score is

$$Z_m = f_0^m(x_0) + \sum_{t < T} B_t(Q_t(f_{t+1}^m; X_t^m, \pi_t^m) - f_t^m(X_t^m)) + B_T(g(X_T^m) - f_T^m(X_T^m)).$$

Taking expectations cancels the fitted values at every date, including both endpoints. The score has expected value equal to actual policy cost even if the networks are poor approximations. To obtain bounded support, write the centered summand as the nonnegative selected-action gap minus the signed Bellman residual. It lies in $[-u_t, \eta_t - \ell_t]$. The terminal term lies in $[-u_T, -\ell_T]$. Thus the score interval has width precisely the corresponding ideal policy bound. Subtracting two scores has a support width equal to the sum of their policy bounds, but its expectation is the direct difference of policy costs. The distinction is that the scores are evaluated along the policies' trajectories; two regret upper bounds are never used as a surrogate observation of the cost difference.

5.2. *Continuous innovations through uniform bins*

For each comparison, draw independent integer arrays K_{itj} uniformly in $\{0, \dots, 2^{40} - 1\}$. A bin represents the continuous interval

$$\left[\frac{1}{32} \left(\frac{2K_{itj}}{2^{40}} - 1 \right), \frac{1}{32} \left(\frac{2(K_{itj} + 1)}{2^{40}} - 1 \right) \right].$$

The exact uniform innovation law can be represented by a uniform bin followed by a uniform draw inside it. The implementation encloses every draw in that bin and needs no additional within-bin random number. Both policies use the same innovation bins but propagate their own states and choose their own actions.

At each date and in each coordinate, the actor is the half-up nearest-node rule. If a state interval meets more than one cell, the implementation includes all eligible node actions and propagates their componentwise hull. Such a hull need not itself be a feasible action, but interval evaluation over the hull still encloses every feasible action actually selected; it is used for enclosure, not as a new policy. The bounding arithmetic covers network values, analytic expectations, transition, stage cost, and terminal cost. Intersecting sample score enclosures with the deterministic support is valid for every bin.

For a bin K_i , the computed endpoints A_i, D_i are deterministic functions with $A_i \leq Z_i \leq D_i$ for every within-bin realization. Hence $\mathbb{E}A_i \leq \mathbb{E}Z_i \leq \mathbb{E}D_i$. Independence of bin draws makes each endpoint sequence independent and identically distributed, conditional on the frozen policies. Endpoint widening changes numerical precision, not the underlying innovation law.

5.3. Simultaneous coverage

Let $q = 24$, $\alpha = 0.01$, and $n = 65536$. Scale each endpoint sequence to its deterministic support of width B . Apply the one-sided empirical Bernstein inequality of [Maurer and Pontil \(2009\)](#), Theorem 4, to the lower sequence with reversed sign and to the upper sequence in its usual direction, each at tail probability $\alpha / (2q)$. Rescaling gives the radius in the main article with logarithm $\log(4q/\alpha) = \log(9600)$. The union bound over all 48 tails yields simultaneous coverage at least 0.99. Dependence between comparisons is not excluded or required to vanish; only within-comparison trajectory independence enters this argument.

The code computes means by outward pairwise summation. If $[\underline{m}, \overline{m}]$ contains the true sample mean, interval squares of $x_i - [\underline{m}, \overline{m}]$ bound every squared deviation from that mean. Their outward sum divided by $n - 1$ therefore bounds sample variance above. The logarithm uses the inherited interval series. A proposed floating

square root is checked by exact rational squaring and enlarged until it is an upper bound. Since the radius is increasing in variance, range, and logarithm, these substitutions preserve coverage.

This is a probability theorem for the stipulated independent simulation model. The fixed PCG64 seed identifies the reproducible realization; it is not a mathematical proof that a deterministic sequence is independent. The comparison states, policies, sample count, and multiplicity are fixed before the new simulations. No optional stopping or policy selection is performed. All 24 intervals contain zero, so no sign or equivalence conclusion is made.

TABLE I: All direct cost intervals. Endpoints are multiplied by 10^4 . States are ordered as in the main article.

Price	Training seed	State	Lower	Upper
1	104729	1	-1.302334	1.319566
1	104729	2	-1.343033	1.331605
1	104729	3	-1.272277	1.395933
1	104729	4	-1.318164	1.380020
1	130363	1	-0.913721	1.096187
1	130363	2	-0.930282	1.016606
1	130363	3	-0.877286	1.070411
1	130363	4	-0.930738	1.031036
1	155921	1	-1.119909	1.057248
1	155921	2	-1.155572	1.015951
1	155921	3	-1.119496	1.035950
1	155921	4	-1.182758	1.039751
4	104729	1	-1.110817	0.892852
4	104729	2	-1.089372	0.876343
4	104729	3	-1.101773	0.870680
4	104729	4	-1.301186	0.663547
4	130363	1	-1.217968	1.253187
4	130363	2	-1.218284	1.248764
4	130363	3	-1.219101	1.252803

Price	Training seed	State	Lower	Upper
4	130363	4	-1.227610	1.207526
4	155921	1	-1.192728	1.201462
4	155921	2	-1.190404	1.185857
4	155921	3	-1.176320	1.205789
4	155921	4	-1.171505	1.216969

6. WORK ACCOUNTS, COMPONENT DIAGNOSTICS, AND REPRODUCTION

TABLE II

ADDITIONAL VERIFICATION WORK AND LOCAL DIAGNOSTIC CLOCKS

Service	Mesh	Operations	Retained actor	Seconds
041	4096	50,393,100	8,196	11.920
089	4096	50,393,100	8,196	67.576
120	4096	50,393,100	8,196	59.052
124	4096	50,393,100	8,196	14.998
128	256	2,054,388,096	133,128	140.429
145	4096	50,393,100	8,196	87.250
147	4096	50,393,100	8,196	15.473

Operations count Bellman transitions for scalar services and neuron expectations for the coupled service; these are different units, not cross-method speed estimates. The additional verification clocks include computations of fine-grid action proposals even though those proposals are not deployed. They do not repeat training or reconstruct the original service clock. Jobs were scheduled concurrently on a resource-limited local host; these clocks describe those diagnostics and cannot be read as isolated cross-method speed measurements.

The original records expose state coverage, action coverage, nodal gaps, and terminal bounds. A representation/optimizer decomposition is not identifiable from them: different parameter errors and fitting trajectories can produce the same

signed Bellman residual. We report the available components as an error account, not a causal identification of each source of approximation error. Training residuals are available as separate diagnostics and are not added twice to a certificate that already evaluates the trained functions.

A fresh reproduction obtains the two pinned input artifacts, verifies their archive digests, and uses a clean result directory. The study commands deliberately refuse to overwrite completed records. The deterministic audit reconstructs target counts and signed comparisons from all original records, checks retained network and actor hashes, and reconstructs every new confidence endpoint from stored moments and support. Regression tests exercise interval arithmetic, exact finite-state policy inequalities, offset invariance, and source identities. The release manifest binds every active source, table, result, and compiled document.

7. RELATIONSHIP TO THE RETAINED THEORY

The full controlled-diffusion and viscosity analysis, economic boundary and transversality conditions, recursive utility and endogenous-preference models, temporal selves, games, quadratic hidden-factor construction, numerical acceptance theorems, and historical experiments remain in their unchanged source files and compiled companions. The new main article does not purport to re-prove them by citing the nonlinear experiment. The preservation map records file hashes and LaTeX labels, and the response gives the location of each current answer to the referee.

The current revision supplies an integrated paper and new proofs and evidence. The new constructive catalogue supplies a separate horizon/price study and repeated isolated timings for its declared backend. An adaptive-grid comparator, a nonlinear dimension frontier, and repeated isolated comparisons for the original optimizers remain distinct evidence requirements; retained-policy and paired-cost results are not substitutes for them.

8. CONSTRUCTIVE NEURAL POLICIES FROM PRIMITIVE MODULI

This section supplies the complete implementation-independent proof of the constructive theorem in the main article. Its assumptions are a compact state set $X \subset \mathbb{R}^d$,

a common compact action set $A \subset \mathbb{R}^a$, a finite horizon, nonnegative discount factors, an invariant transition, a common innovation law, and state/action-independent monotone cash-invariant conditional evaluation on a common bounded-function domain. The primitive moduli are those in the main article's constructive section. Effective compactness and finite certified numerical evaluation are used only for finite termination, not for the analytic inequalities.

8.1. Bellman moduli and labels

If $\|u - v\|_\infty \leq r$, then $v - r \leq u \leq v + r$. Monotonicity and cash invariance give $|\rho_t(u) - \rho_t(v)| \leq r$. Consequently, when f_{t+1} is L_{t+1} -Lipschitz,

$$|Q_t(f_{t+1}; x, a) - Q_t(f_{t+1}; y, b)| \leq L_t \|x - y\|_1 + L_t^a \|a - b\|_1,$$

where $L_t = C_{x,t} + \beta_t F_{x,t} L_{t+1}$ and $L_t^a = C_{a,t} + \beta_t F_{a,t} L_{t+1}$. Infima over the same action set preserve the state modulus. Compactness and continuity give minima. Alternatively, the same argument uses arbitrarily close minimizers and then takes their error to zero.

Let $H_t = T_t f_{t+1}$, $q_{ij} = Q_t(f_{t+1}; x_i, a_j)$, and $|\hat{q}_{ij} - q_{ij}| \leq e$. The state and action grids have radii h, k . Since every grid action is feasible,

$$y_i = \min_j \hat{q}_{ij} \geq \min_j q_{ij} - e \geq H_t(x_i) - e.$$

Choose an action minimizer a^* and a grid action within k of it. Then

$$y_i \leq q_{ij^*} + e \leq H_t(x_i) + L^a k + e.$$

These are bounds on the label array actually produced by the own-future recursion. They do not use V_{t+1}^* and do not assume that a regression loss is small.

8.2. Envelope, residual, and selected action

For arbitrary finite labels y_i , define $E_y(x) = \min_i \{y_i + L\|x - x_i\|_1\}$. Each cone is L -Lipschitz. For a minimizing cone at x' , $E_y(x) - E_y(x') \leq L\|x - x'\|_1$, and reversing the points proves the Lipschitz assertion. If $\max_i |y_i - y'_i| \leq \delta$, then every cone shifts

by at most δ , so $\|E_y - E_{y'}\|_\infty \leq \delta$. In particular, numerical label errors cannot inflate the certified slope.

For every node, $y_i + L\|x - x_i\|_1 \geq H_t(x_i) - e + L\|x - x_i\|_1 \geq H_t(x) - e$. Taking the minimum proves the lower residual bound. Choosing a nearest node $i(x)$ gives

$$E_y(x) \leq H_t(x_{i(x)}) + L^a k + e + Lh \leq H_t(x) + L^a k + e + 2Lh.$$

If $j(i)$ minimizes the numerical queries, then

$$q_{i,j(i)} \leq \hat{q}_{i,j(i)} + e = y_i + e \leq H_t(x_i) + L^a k + 2e.$$

The difference of the chosen-action value and H_t has state modulus at most $2L$. Thus the nearest-node actor has gap at most $L^a k + 2e + 2Lh$. A deterministic rule for ties among finitely many closed Voronoi cells gives a Borel actor. Any tied nearest node satisfies the same inequality. Because the action set is common, this actor is feasible at every state. This proof would not establish feasibility for an unrestricted state-dependent correspondence.

At the terminal date, $H_T = g$ and there is no action minimization. Labels within e_T of g yield $-e_T \leq f_T - g \leq e_T + 2L_g h_T$. Starting from the terminal network and proceeding backward establishes the assumed continuation moduli at every date. The policy theorem, proved earlier in this supplement, then yields

$$J_0^\pi(x) - V_0(x) \leq \sum_{t < T} B_t(2L_t^a k_t + 4e_t + 4L_t h_t) + B_T(2e_T + 2L_g h_T).$$

The comparison is with the optimum over all admissible adapted policies, not just policies constant on the grid. The action covering allowance is precisely what makes the continuous-action comparison possible.

8.3. Realization and finite termination

An absolute value uses two ReLU gates. A sum of d absolute values with affine coefficients computes one cone. The identity $\min(u, v) = u - \max(u - v, 0)$ computes pairwise minima. A balanced finite tree consumes K cones in $O(\log K)$ stages, with $O(K)$ additional gates. Positive and negative channels represent signed intermediate scalars in an ordinary feed-forward realization. The total number of coefficients

is $O(dK)$; a bounded-fan-in realization can replace each affine sum by a logarithmic-depth addition tree. This is an explicit network with weights determined by labels, state nodes, and the primitive upper modulus.

All constants may be replaced by computable rational upper bounds. The labels can be rational rounded midpoints of certified numerical intervals. State/action covers must be supplied with verifiable covering and feasibility radii, rather than presumed because a set is topologically compact. For the budget $s = \varepsilon / \sum B_t$, the nonterminal terms satisfy

$$2(s/8) + 4(s/8) + 4(s/16) = s,$$

and terminal terms satisfy $2(s/4) + 2(s/4) = s$. Positive-radius covers have finitely many nodes and numerical queries reach the specified tolerances by hypothesis. Backward construction therefore takes finite work and the policy loss is at most ε . This is a deterministic algorithm theorem. It does not turn a finite implementation cap into a termination guarantee, and it does not provide a convergence theorem for Adam, stochastic gradient descent, or an arbitrary nonconvex loss.

The work expression in the article counts all state–action–innovation evaluations of the already constructed future network. If numerical precision increases with the requested tolerance, the bit cost of each operation and of each weight must also increase in the account. A tensor cover retains its exponential state/action dimension factors. Analytic or sparse integration can reduce the innovation work but does not remove the covering factors merely by being called neural computation.

8.4. *Exact scalar compilation*

Let $0 = x_0 < \dots < x_N = 1$ and let the labels be arbitrary rational numbers. Define

$$z_i = \min_j \{y_j + L|x_i - x_j|\}.$$

A left-to-right pass computes the minimum over $j \leq i$ and a right-to-left pass incorporates $j \geq i$. The result is the greatest L -Lipschitz vector lying below the labels: any such vector w satisfies $w_i \leq w_j + L|x_i - x_j| \leq y_j + L|x_i - x_j|$ for every j . In particular $|z_{i+1} - z_i| \leq L(x_{i+1} - x_i)$.

For $x \in [x_i, x_{i+1}]$, every cone centered to the left is represented by the best left intercept and every cone centered to the right by the best right intercept. Therefore

$$E_y(x) = \min\{z_i + L(x - x_i), z_{i+1} + L(x_{i+1} - x)\}.$$

For $L > 0$, the intersection is

$$r_i = \frac{x_i + x_{i+1}}{2} + \frac{z_{i+1} - z_i}{2L} \in [x_i, x_{i+1}].$$

When the intersection is interior it is the only additional knot; endpoint intersections cause no duplicate knot. When $L = 0$ the envelope is constant. This proves equality of the compiled piecewise-linear function and the defining network everywhere, including non-dyadic intersections and ties.

For rational knots r_j , values v_j , and slopes s_j , the antiderivative on one cell is

$$A(x) = A(r_j) + v_j(x - r_j) + \frac{1}{2}s_j(x - r_j)^2.$$

Rational trapezoidal accumulation computes $A(r_j)$ exactly. Under a continuous uniform innovation on $[-\sigma, \sigma]$, the own-future expectation is $[A(b + \sigma) - A(b - \sigma)] / (2\sigma)$. This is an identity under the original law, not a finite-bin change of the economy.

The numerical evaluator encloses each rational coefficient and each native arithmetic operation outward. At a query near a rounded non-dyadic knot, it resolves the cell using the exact rational knot and the exact rational value of the binary64 query. It does not use a rounded knot as a silent change of the partition. An uncertain antiderivative argument is evaluated at its midpoint and expanded by $\sup |f|$ times its radius; an uncertain innovation center is expanded by $\text{Lip}(f)$ times its radius. Intersections with $[0, 1]$ discard only rounding overhang when the exact primitive argument is known to lie in that invariant interval. These steps separate analytic integration from arithmetic rounding.

8.5. Matched spline account and measurement scope

For the spline comparator, linear interpolation of labels with node errors between $-e$ and $L^a k + e$ gives a residual interval $[-e - Lh, L^a k + e + Lh]$. Indeed the inter-

polating weights are nonnegative and sum to one, and their weighted distance to the query point is at most h on the uniform one-dimensional mesh. Its width is $L^a k + 2e + 2Lh$, the same form as the envelope width. The nearest-node actor allowance is also the same form. The spline's actual exact slope, not the neural primitive upper bound, is propagated into its next date's query modulus. Both terminal certificates conservatively use the primitive terminal modulus. Neither comparator borrows the other's targets.

The catalogue records exact rational gap formulas, upper binary64 endpoints, all interval-error allowances, defining labels, compiled nodes, and selected actions for every rung. It also records evaluated Bellman queries, continuous-law integrals, antiderivative queries, binary-search upper counts, deployed actor scalars, rational storage quantities, maximum stored numerator/denominator bit lengths, checkpoint bytes, process peak resident memory, internal clocks, and process clocks. Primitive-object counts are not described as a complete floating-point or bit-operation count. Peak memory includes the interpreter and warm-up. The internal target clock includes failed rungs and durable checkpoint output; independent economic comparison is absent rather than costless. These scope qualifications are part of the empirical estimand.

A first-rung timing repetition is not a new random economic environment. For each fixed method/horizon/price combination, the three exact certificate arrays and deployed objects should coincide; their clocks need not. Equality of object hashes is audited separately from timing dispersion. This supplies reproducibility for the declared deterministic generator without inferring a probability of success for another training algorithm.

9. COMPLETE CONSTRUCTIVE VERIFICATION FRONTIERS

TABLE III: All 48 distinct method–economy–resolution
rows

Generator	T	p	N	Gap	Prefix queries	Median s
ReLU envelope	2	1	16	1.96222	578	0.0104
ReLU envelope	2	1	32	0.98111	2,756	0.0252
ReLU envelope	2	1	64	0.49056	11,206	0.0534
ReLU envelope	2	1	128	0.24528	44,488	0.1200
ReLU envelope	2	1	256	0.12264	176,586	0.3009
ReLU envelope	2	1	512	0.06132	702,924	0.8474
Spline	2	1	16	1.74262	578	0.0072
Spline	2	1	32	0.88564	2,756	0.0167
Spline	2	1	64	0.44641	11,206	0.0345
Spline	2	1	128	0.22410	44,488	0.0800
Spline	2	1	256	0.11228	176,586	0.2144
Spline	2	1	512	0.05619	702,924	0.6777
ReLU envelope	2	4	16	2.00763	578	0.0101
ReLU envelope	2	4	32	1.00382	2,756	0.0248
ReLU envelope	2	4	64	0.50191	11,206	0.0529
ReLU envelope	2	4	128	0.25095	44,488	0.1192
ReLU envelope	2	4	256	0.12548	176,586	0.2944
ReLU envelope	2	4	512	0.06274	702,924	0.8450
Spline	2	4	16	1.80513	578	0.0075
Spline	2	4	32	0.91577	2,756	0.0168
Spline	2	4	64	0.46103	11,206	0.0345
Spline	2	4	128	0.23136	44,488	0.0799
Spline	2	4	256	0.11587	176,586	0.2151
Spline	2	4	512	0.05799	702,924	0.6740
ReLU envelope	4	1	16	3.91560	1,156	0.0174
ReLU envelope	4	1	32	1.95780	5,512	0.0430
ReLU envelope	4	1	64	0.97890	22,412	0.0927

Generator	T	p	N	Gap	Prefix queries	Median s
ReLU envelope	4	1	128	0.48945	88,976	0.2132
ReLU envelope	4	1	256	0.24473	353,172	0.5335
ReLU envelope	4	1	512	0.12236	1,405,848	1.5540
Spline	4	1	16	2.99898	1,156	0.0127
Spline	4	1	32	1.52738	5,512	0.0293
Spline	4	1	64	0.77070	22,412	0.0618
Spline	4	1	128	0.38711	88,976	0.1455
Spline	4	1	256	0.19399	353,172	0.4028
Spline	4	1	512	0.09711	1,405,848	1.2654
ReLU envelope	4	4	16	4.00092	1,156	0.0171
ReLU envelope	4	4	32	2.00046	5,512	0.0424
ReLU envelope	4	4	64	1.00023	22,412	0.0931
ReLU envelope	4	4	128	0.50012	88,976	0.2125
ReLU envelope	4	4	256	0.25006	353,172	0.5349
ReLU envelope	4	4	512	0.12503	1,405,848	1.5495
Spline	4	4	16	3.17460	1,156	0.0128
Spline	4	4	32	1.61224	5,512	0.0293
Spline	4	4	64	0.81227	22,412	0.0612
Spline	4	4	128	0.40770	88,976	0.1436
Spline	4	4	256	0.20423	353,172	0.4038
Spline	4	4	512	0.10221	1,405,848	1.2705

Every row represents three byte-identical policy and certificate checkpoints with distinct clocks. Complete counts, storage, memory, and the individual clocks are deposited in the machine-readable records. The full frontier includes unsuccessful rungs; no refined neural row is substituted for an original failed service.

10. WITNESS-PRESERVING POLICY CONSTRUCTION

This section establishes the additional policy sandwich and the witness-preserving implementation. It uses the same primitive economy, admissible-policy

comparison and common evaluation domain as the corresponding main-text results. All inequalities concern the continuation actually constructed from its own fitted future.

10.1. Order comparison and signed accounts

Suppose $f_t - \mathcal{T}_t f_{t+1} \leq u_t$ and $\mathcal{T}_t^\pi f_{t+1} - f_t \leq d_t$. Define

$$U_t = \sum_{j=t}^{T-1} \left(\prod_{k=t}^{j-1} \beta_k \right) u_j + \left(\prod_{k=t}^{T-1} \beta_k \right) u_T,$$

with empty products equal to one. The terminal inequality gives $V_T^* = g \geq f_T - U_T$.

If $V_{t+1}^* \geq f_{t+1} - U_{t+1}$, monotonicity and cash invariance yield

$$V_t^* = \mathcal{T}_t V_{t+1}^* \geq \mathcal{T}_t f_{t+1} - \beta_t U_{t+1} \geq f_t - u_t - \beta_t U_{t+1} = f_t - U_t.$$

The same induction for the fixed-policy map, beginning at $g \leq f_T - \ell_T$, gives

$$J_0^\pi \leq f_0 + \sum_{t < T} B_t d_t - B_T \ell_T.$$

Subtracting proves the one-sided sandwich. Neither u_t nor d_t is replaced by an absolute value; their signs cancel the effect of arbitrary date-specific changes of origin. The lower loss bound follows from Bellman optimality against all admissible adapted policies. This argument uses no independence of simulations and no linearity of the recursive functional. It is distinct from the expectation-specific paired telescoping theorem retained earlier in this supplement.

The old centered certificate is the special bound $d_t \leq \eta_t - \ell_t$. Its possible slack is due to taking two separate extrema of quantities that share the same fitted value. The new witness result controls their sum directly.

10.2. Action transport through the minimum circuit

At a node x_i , the numerical action query attaining y_i corresponds to an actual feasible action a_i . The query enclosure gives $Q(x_i, a_i) \leq y_i + e$. Uniform state Lipschitz continuity of $Q(\cdot, a)$ therefore implies

$$Q(x, a_i) - f(x) \leq e + y_i + L \|x - x_i\|_1 - f(x)$$

for every node and every state. Selecting an index attaining the minimum cone makes the last three terms sum to zero. Finite minima attain their values, and a fixed ordering of the finite indices defines a Borel selector. The common action set is essential: the proof does not transport an action into a state at which that action is infeasible.

Combining $d_t = e_t$ with $u_t = L_t^a k_t + e_t + 2L_t h_t$ proves the displayed witness gap. It also implies the separated actor allowance

$$Q_t(f_{t+1}; x, \pi_t^w(x)) - \mathcal{T}_t f_{t+1}(x) \leq e_t + u_t = L_t^a k_t + 2e_t + 2L_t h_t.$$

Thus the earlier numerical separated budget is also a valid, though weaker, bound for the new witness actor. Using it as an alternative does not substitute the old nearest-node policy. At each nonterminal date, the new sufficient resource choices obey $s/4 + 2(s/4) + 2(s/8) = s$. The terminal bracket is also at most s . Effective finite covers and certified numerical evaluation then give finite termination.

Only the critic is an affine-ReLU circuit. Returning a comparison index and an attached action is a finite measurable algorithm with a possibly discontinuous policy. Classical Lipschitz extension and elementary minimum gates remain credited in the main article. The result does not assert their novelty or claim that an equivalent conventional envelope implementation is impossible.

10.3. *Approximate indices and numerical deployment*

Let x' be an acquired state with $\|x - x'\|_1 \leq r$. Suppose the selected index satisfies

$$y_i + L\|x' - x_i\|_1 \leq f(x') + \nu.$$

The cone and the envelope are both L -Lipschitz. Hence

$$y_i + L\|x - x_i\|_1 - f(x) \leq \nu + 2Lr.$$

A feasible executed action within distance α of a_i adds at most $L^a \alpha$ to Q . This proves the stated execution allowance. Rounding a rational switching boundary without a checked comparison can change the selected index; it must be covered by ν or by a certified interval containing all eligible indices. An exact tie presents no difficulty,

since both attaining cones give the same certified upper bound. The current records store exact rational boundaries and define the ideal selector. They do not claim a measured deployment performance for arbitrary hardware.

10.4. *Scalar compilation with original witnesses*

Let $0 = x_0 < \dots < x_N = 1$ and let the original numerical labels y_i be arbitrary rational numbers. Define pairs

$$(p_i, l_i) = \min_{j \leq i} (y_j + L(x_i - x_j), j), \quad (q_i, r_i) = \min_{j \geq i} (y_j + L(x_j - x_i), j),$$

where pairs are ordered first by value and then by index. A forward pass computes the first array by comparing (y_i, i) with $(p_{i-1} + L(x_i - x_{i-1}), l_{i-1})$; a backward pass computes the second. The index always refers to an original label and its action, not to a newly regularized value with no action witness.

On $[x_i, x_{i+1}]$, the envelope is the minimum of

$$p_i + L(x - x_i) \quad \text{and} \quad q_{i+1} + L(x_{i+1} - x),$$

with witnesses l_i and r_{i+1} . When $L > 0$ their intersection is

$$s_i = \frac{x_i + x_{i+1}}{2} + \frac{q_{i+1} - p_i}{2L}.$$

Unlike an intersection computed after closing both labels, this intersection may lie outside the cell. In that case one affine candidate attains the minimum throughout the cell. An interior intersection divides the cell into exactly two witness regions. If $L = 0$, compare the two constants directly. The implementation preserves every original node boundary and every interior switch, and chooses the right cell at an exact boundary. Continuity ensures that an adjoining witness remains attaining at the boundary. Thus there are at most $2N$ witness intervals, with rational endpoints and original action indices.

Once rational endpoints are loaded, an ordered search locates a region using a logarithmic number of comparisons. Reading and parsing all stored rational cuts anew costs linear work and is not removed by that comparison bound. The reference accessor used by the tests performs that parsing; the catalogue measures

construction and serialization, not repeated policy-query throughput. The record's binary-search count is only a count of ordered comparisons, not the complete cost of that accessor. Rational numerator/denominator lengths, cuts, witness indices and original actions are stored separately.

10.5. *A common strengthened nearest-node certificate*

For a nearest-node policy and any fitted piecewise-affine continuation, let

$$D = \max_i \sup_{x \in C_i} \{y_i + L|x - x_i| - f(x)\}.$$

The selected-node query and the state modulus give $Q(x, \pi(x)) - f(x) \leq e + D$. On each intersection of a nearest-node cell with a linearity interval of f , divided also at x_i , the expression is affine. Its maximum is attained at an endpoint. Consequently it suffices to enumerate the union of the fitted knots, original state nodes and nearest-cell boundaries. At a shared nearest-cell boundary both possible node indices are included. This is an exact all-state calculation, rather than a sampled maximum.

Both the cone-nearest ablation and spline-nearest comparator use this calculation and their own valid optimal-residual upper bounds. The former has $D \geq 0$ because the cone envelope lies below every candidate cone. Its witness counterpart achieves zero for this particular label-transport upper account. This statement is about the account, not an optimality theorem for the actual policy among all feasible controls. The spline's continuation is not replaced by the cone envelope and continues to propagate its own exact Lipschitz constant.

10.6. *Evidence identities and scope of reproduction*

The protocol predates the complete 36-service execution. Fourteen new checks examine exact compilation, noisy and dominated labels, ties, finite-policy comparison, shift cancellation, common nearest-cell bounds, exact continuous-uniform integration in small examples, and immutable numerical dependencies. They supplement the analytic proof rather than replace it. The source hashes, all 216 checkpoints, and complete process clocks are retained. Formula reconstruction verifies each one-sided component, every first crossing, and the identity of the two neural critics in

the actor ablation. Three repetitions of a deterministic policy are not three economic draws.

The latest R43 supplemental report concerns a different, earlier incomplete snapshot. The current reply does not claim that its missing source parts were present. It identifies the complete R46 source tree as the new review object and retains the earlier review and incomplete historical branch unchanged. The predeclared R46 execution, R45 construction catalogue, R44 postselected diagnostics, original R42 capped services and original paired cost comparisons remain distinct evidence objects.

TABLE IV: Complete matched witness frontiers

T	p	N	Witness	Nearest neural	Spline
2	1	16	1.097841	1.558158	1.222038
2	1	32	0.548921	0.790405	0.622981
2	1	64	0.274460	0.398115	0.314494
2	1	128	0.137230	0.199802	0.158000
2	1	256	0.068615	0.100089	0.079188
2	1	512	0.034308	0.050092	0.039641
2	4	16	1.120546	1.615348	1.269376
2	4	32	0.560273	0.815732	0.645500
2	4	64	0.280137	0.411198	0.325309
2	4	128	0.140068	0.206040	0.163341
2	4	256	0.070034	0.103215	0.081831
2	4	512	0.035017	0.051637	0.040957
4	1	16	2.060395	2.916323	2.035153
4	1	32	1.030197	1.478057	1.040370
4	1	64	0.515099	0.744163	0.525918
4	1	128	0.257549	0.373386	0.264396
4	1	256	0.128775	0.187021	0.132558
4	1	512	0.064387	0.093593	0.066369

T	p	N	Witness	Nearest neural	Spline
4	4	16	2.103056	3.034798	2.161469
4	4	32	1.051528	1.533865	1.100879
4	4	64	0.525764	0.772861	0.555422
4	4	128	0.262882	0.387323	0.278981
4	4	256	0.131441	0.193954	0.139801
4	4	512	0.065720	0.097039	0.069979

TABLE V: Full-frontier isolated construction clocks in seconds

Method	(T, p)	Median	Minimum	Maximum	Peak MiB
cone-witness	(2, 1)	0.9631	0.9595	0.9817	113.8
cone-witness	(2, 4)	0.9596	0.9583	0.9709	113.9
cone-witness	(4, 1)	1.8000	1.7928	1.8025	116.4
cone-witness	(4, 4)	1.7456	1.7104	1.7899	116.4
cone-nearest	(2, 1)	1.1797	1.1739	1.2017	113.7
cone-nearest	(2, 4)	1.1798	1.1691	2.1261	113.7
cone-nearest	(4, 1)	2.2305	2.1667	2.2802	115.4
cone-nearest	(4, 4)	2.2163	2.2158	2.2221	115.4
spline-nearest	(2, 1)	0.8632	0.8424	0.8706	113.1
spline-nearest	(2, 4)	0.8755	0.8725	0.8964	113.0
spline-nearest	(4, 1)	1.6690	1.6628	1.6888	115.1
spline-nearest	(4, 4)	1.6707	1.5893	1.6783	115.1

Clocks include all six rungs through checkpoint fsync. Peak resident memory includes the interpreter and warm-up. These are three local repetitions, not independent economic draws.

11. FEASIBLE TRANSPORT UNDER ENDOGENOUS CONSTRAINTS

We give the details underlying the feasible-witness transport theorem in the main article. All norms below are the specified ℓ^1 norms. The evaluation functional is the same at the compared state–action pairs. Its monotonicity and cash invariance imply that changing its bounded random input by at most b changes its value by at most b . Thus the primitive state and action inequalities on the feasible graph imply

$$|Q_t(f_{t+1}; x, a) - Q_t(f_{t+1}; z, b)| \leq A_t^Q \|x - z\|_1 + D_t^Q \|a - b\|_1$$

whenever $a \in A_t(x)$ and $b \in A_t(z)$. The proof never evaluates Q_t outside that graph.

11.1. The Bellman modulus and noisy feasible labels

Fix $a \in A_t(x)$. Compactness and the Hausdorff condition give $b \in A_t(z)$ with $\|a - b\|_1 \leq K_t^A \|x - z\|_1$. It follows that

$$\mathcal{T}_t f_{t+1}(z) \leq Q_t(f_{t+1}; x, a) + L_t \|x - z\|_1.$$

Taking the infimum over a and then reversing x, z establishes the claimed Bellman modulus, without assuming that the minimizer is known. At node i , the minimum of exact objective values over its feasible action net is at least the continuous feasible infimum and at most that infimum plus $D_t^Q k_t$. Taking the minimum of queries with absolute error at most e_t adds at most e_t in each direction.

For every cone, the lower label inequality and the Bellman modulus give

$$y_{t,i} + L_t \|x - x_{t,i}\|_1 \geq \mathcal{T}_t f_{t+1}(x) - e_t.$$

For a nearest state node, the upper label inequality gives an upper bound $\mathcal{T}_t f_{t+1}(x) + D_t^Q k_t + e_t + 2L_t h_t$. The finite minimum has the same bounds and is L_t -Lipschitz. The next backward step therefore has precisely the modulus required by the construction, independently of the numerical label errors.

11.2. Repair, selection and the dynamic account

The stored action $a_{t,i}$ attains an original numerical query; hence its exact node objective is at most $y_{t,i} + e_t$. For the minimizing cone, the feasible-pair inequality

and repair displacement give

$$Q_t(f_{t+1}; x, \Gamma_{t,i}(x)) - f_t(x) \leq e_t + D_t^Q \zeta_t.$$

A fixed ordering of finitely many continuous cone values gives a Borel index selector. Composing the finitely many Borel repairs with that selector gives a feasible Borel policy. The one-sided sandwich now supplies the discounted loss bound, with the original terminal interval unchanged. The full-policy comparison is exactly the Bellman comparison assumed in the controlled economy; it is not a comparison restricted to the node action arrays.

The four nonterminal resource allocations make the bracket at most $s/4 + s/4 + s/4 + s/4 = s$. The terminal bracket is at most s . Effective finite covers, feasible action nets, query enclosures and repair procedures consequently give finite termination. Where only an approximate repair is computable, its certified displacement is charged through $D_t^Q \zeta_t$ and its actual action must still be feasible. A numerical tolerance for an infeasible constraint violation is not included in this theorem.

11.3. *A transparent exact check*

Let $X = [0, 1]^2$, $b(x) = (x_1 + x_2)/2$, $A(x) = [0, b(x)]$, and consider the one-date objective

$$Q(x, a) = (a - 3/4)^2 + x_1/8 + x_2/16.$$

Here $A^Q = 1/8$, $D^Q = 3/2$, $K^A = 1/2$, and $L = 7/8$ are valid moduli. The exact optimum is attained at $\min\{3/4, b(x)\}$. For a feasible anchor action, clipping at $b(x)$ changes the action by at most $|b(x) - b(x_i)| \leq \|x - x_i\|_1/2$. The transport theorem therefore applies directly, and a reference implementation can compare the exact repaired-policy loss with the analytic bound using rational arithmetic.

The additional tests independently form feasible node nets and noisy labels, compute the cone minimum by direct enumeration, and check the Bellman and selected-policy enclosures on a fixed rational set. They also check the Hausdorff modulus, exact feasibility, approximate-repair accounting, nonminimizing-index allowances and the zero-constraint-variation reduction. The analytic proof above, rather than a finite grid test, establishes the all-state claim. The tests are not training

data, timing observations, or evidence of a high-dimensional speed advantage. No R46 catalogue record is altered by these checks.

12. ACQUIRED STATES AND IMPLEMENTABLE WITNESSES

This section supplies the implementation and statistical details for the new acquisition-aware results. The earlier proofs remain in force on their stated domains. To avoid conflating economic objects, the constrained study and the direct scalar comparison are separate experiments: the former constructs new two-state policies, while the latter evaluates the actual scalar policies previously constructed.

12.1. Causal information and the one-sided comparison

Let $\mathcal{H}_t$ contain the primitive history and the observations available before date- t action. The observation process is causal and preserves the stipulated conditional law, or conditional recursive evaluator, for the next primitive innovation. This condition permits state errors to depend arbitrarily on the available past. It does not permit a sensor to reveal a future innovation without changing the economic information structure. The full-information optimum is over feasible policies adapted to this permitted history. Uniform one-step inequalities therefore apply to observation-dependent actions as well as exact-state Markov selectors.

Suppose $f_t - \mathcal{T}_t f_{t+1} \leq u_t$, and suppose every permitted observation and true state obey $\mathcal{T}_t^{\hat{\pi}} f_{t+1} - f_t \leq d_t$. If $\ell_T \leq f_T - g \leq u_T$, induction gives

$$V_t^* \geq f_t - \sum_{j=t}^{T-1} B_{t,j} u_j - B_{t,T} u_T, \quad J_t^{\hat{\pi}} \leq f_t + \sum_{j=t}^{T-1} B_{t,j} d_j - B_{t,T} \ell_T.$$

The second inequality is interpreted on $\mathcal{H}_t$ when the policy uses the observation history. Monotonicity and cash invariance justify every induction step on the common evaluation domain. Subtraction yields the sum of $u_j + d_j$ and terminal width; no actor continuity is used. A date shift $f_t \mapsto f_t + s_t$ adds $s_t - \beta_t s_{t+1}$ to u_t and subtracts the same quantity from d_t . The policy-loss account is invariant.

At a selected node, retain the original action $a_{t,i}$ and a label whose numerical error is at most e_t . For a feasible deployed action $\hat{a}$, state and action transport give

$$Q_t(f_{t+1}; x, \hat{a}) \leq y_{t,i} + e_t + A_t^Q \|x - x_{t,i}\|_1 + D_t^Q \|\hat{a} - a_{t,i}\|_1.$$

Substitute the robust-repair inequality and $L_t = A_t^Q + D_t^Q K_t^A$. The cone at x is within $L_t r_t$ of its value at the acquired state $\hat{x}$. An index with numerical excess v_t is within v_t of the envelope there. Finally the envelope, being a minimum of L_t -Lipschitz functions, is L_t -Lipschitz, adding another $L_t r_t$. Hence

$$Q_t(f_{t+1}; x, \hat{a}) - f_t(x) \leq e_t + v_t + 2L_t r_t + D_t^Q \zeta_t^{\text{acq}}.$$

Combining this with the optimal-residual upper bound $D_t^Q k_t + e_t + 2L_t h_t$ proves the main theorem. This proof charges an objective excess even when two selected actions differ substantially at a boundary.

For a capacity interval, exact repair at the true state has displacement $(a_i - \kappa(x))_+ \leq K^A \|x - x_i\|_1$. Replacing capacity by a nonnegative certified lower value changes the repaired action by at most $\kappa(x) - \underline{\kappa}(\hat{x})$. Downward quantization adds less than Δ . These inequalities prove both feasibility and the stated displacement bound. In the executed affine-capacity economy the acquisition box has coordinate endpoints $\max(0, \hat{x}_j - \eta)$ and $\min(1, \hat{x}_j + \eta)$. Its minimum capacity is computed at the lower corner. All these quantities are dyadic, so that the capacity and downward action rounding are exact at the specified resolutions.

12.2. Finite-precision indices and the identical envelope

The scalar and two-state experiments store labels on a dyadic lattice. In the acquired-state deployment, nodes and measured coordinates have the declared dyadic denominators; their coordinate differences and the sum of two absolute differences are exactly representable in binary64. No overflow or underflow occurs in the evaluated range. For a cone $v_i = y_i + L \|\hat{x} - x_i\|_1$, let $\tilde{L}$ denote the binary64 slope and $\tilde{v}_i$ its computed score. The standard product-and-sum bound, with unit roundoff $u = 2^{-53}$, gives

$$|\tilde{v}_i - v_i| \leq d|\tilde{L} - L| + \gamma_2(Y + d|\tilde{L}|) = \tau, \quad \gamma_2 = \frac{2u}{1 - 2u}, \quad Y = \max_i |y_i|.$$

If i minimizes the computed scores and j minimizes the exact scores, then $v_i \leq \tilde{v}_i + \tau \leq \tilde{v}_j + \tau \leq v_j + 2\tau$. Thus $\nu = 2\tau$ works regardless of a tie. More general state formats must charge the formation of distances too; that cost is not suppressed by this dyadic example.

An affine-ReLU minimum gate is $a - (a - b)_+$. For exact inputs it equals the native minimum. Attaching the same index and the same tie rule proves equality of the witness actor, including feasible repair. The proof is algebraic and covers all states, not only the measurement grid. The scalar implementation's compiled right-cell tie convention is held fixed in this control.

Ordinary rounded ReLU gates need not agree bit for bit with native minima. If the exact intermediate minima have absolute value at most M , input errors are at most E , and two subtractions implement one gate, a conservative next-level bound is

$$E_{\text{next}} \leq E + 6u(M + E)$$

under the preceding finite-range conditions. Indeed the first subtraction contributes at most $2u(M + E)$, the positive-part map is nonexpansive, and the last subtraction contributes at most $u(3 + 2u)(M + E)$. The exact minimum is nonexpansive in its two arguments. Iteration from the leaf conversion bound controls a balanced circuit. The retained diagnostic checks all 24 scalar witness checkpoints on 1,025 dyadic points against this budget. Its gate-evaluation clock excludes construction and is not a method-level speed comparison. The native representation and the ReLU representation share parameters; they are not counted as independent trained policies.

13. EXPLICIT CONSTRAINED PRIMITIVES AND RESOURCE ALLOCATION

For the new two-state economy, the action lies in $[0, 1/4]$. Bounding the non-negative state terms separately gives $F_1 \in [1/32, 29/32]$ and $F_2 \in [1/32, 27/32]$. These bounds already establish invariance before any numerical enclosure is intersected with the domain. The two state-column sums of the transition Jacobian are $11/16 - x_2/8$ and $11/16 - x_1/8$; every term used in these sums is nonnegative. The induced ℓ^1 state modulus is therefore $11/16$. The action column has

norm $1/2 + 1/4 = 3/4$. Capacity has gradient $(1/16, 1/16)$ and hence ℓ^1 Lipschitz constant $1/16$.

Let $s = (1/2 - x_1 - x_2)_+$. The first state derivative of stage cost is $2(x_1 - 5/8) + (x_1 - x_2)/2 - 4s$; the second is the symmetric expression. On each of the two short-age regions these derivatives are affine. At the vertices of the resulting polygons their absolute values are at most $13/4$. Doubling the squared-target part changes this bound to $9/2$ for terminal cost. The action derivative $2pa + 16a^3$ is nonnegative and at most $p/2 + 1/4$. These are constants for the primitive equations, not estimates from the evaluation grid.

For a uniform tensor grid, each coordinate is within $1/(2N)$ of a node, giving $h = 1/N$. The maximum feasible interval length is $1/4$, so a uniform N -subdivision feasible net has radius at most $1/(8N)$. At different states the capacity intervals have Hausdorff distance at most $(1/16)\|x - x'\|_1$. The map $a \mapsto \min\{a, \kappa(x)\}$ is measurable and feasible, with the displacement bound proved above. Thus neither an unspecified covering oracle nor an unspecified feasible projection is used here.

A shock cell has length $1/(16M)$. Its conditional mean distance to the midpoint is $1/(64M)$. The transition's shock derivative is $(1, -1)$, of ℓ^1 norm two. Consequently a future continuation of modulus L has midpoint expectation error at most $L/(32M)$. The program encloses numerical evaluation at all midpoints, averages outward, adds this continuous-law remainder, and discounts it. Labels are dyadic rounded centers with an error enclosing the full numerical query. The economic law remains continuous; this is not a proof for a substituted finite-state shock process.

13.1. The strengthened bilinear comparator

Write $\lambda_i(x)$ for the four bilinear interpolation weights in a tensor cell. They are nonnegative, sum to one and reproduce the two coordinates. In one coordinate, at relative position $q \in [0, 1]$, the weighted distance to the two endpoints is $2q(1 - q)/N \leq 1/(2N)$. Adding coordinates gives $\sum_i \lambda_i(x)\|x - x_i\|_1 \leq 1/N$. If the nodal Bellman labels have lower error $-e_t$ and upper error $D_t^Q k_t + e_t$, the graph-Lipschitz Bellman target therefore obeys

$$-e_t - L_t/N \leq f_t(x) - \mathcal{T}_t f_{t+1}(x) \leq D_t^Q k_t + e_t + L_t/N.$$

The interpolant's own Lipschitz modulus is N times the largest adjacent nodal difference in absolute value. Each partial derivative is a convex combination of two adjacent differences. In this execution the differences are dyadic with sufficiently few significant bits to be represented exactly; the computed modulus is consequently an upper bound without a hidden downward rounding.

For the nearest-node actor, each closed Voronoi cell is split at its node coordinates. On each of the at most four resulting rectangles the absolute-distance term is affine, and the continuation is a single bilinear polynomial. Fixing one coordinate makes the difference affine in the other. Successively maximizing both coordinates therefore attains a maximum at a vertex. The union contains at most nine points. Evaluating every point outward bounds the complete supremum, including boundary ties. This proves the selected-policy allowance used by the comparator rather than comparing a sharpened witness bound to an unnecessarily separated baseline.

13.2. *Sufficient caps versus executed caps*

For the cone generator,

$$L_t = C_x + K^A C_a + \beta(F_x + K^A F_a)L_{t+1}.$$

Here $b = \beta(F_x + K^A F_a) = 705/1024 < 1$, giving the horizon-uniform bound $\bar{L} = \max\{L_g, (C_x + K^A C_a)/(1 - b)\}$. The action modulus is at most $\bar{D} = C_a + \beta F_a \bar{L}$. With $M = N$, nonterminal label error is at most $\beta \bar{L}/(32N) + \xi$. Substituting into the one-sided loss bound yields exactly

$$\mathcal{G}_N \leq \frac{S_T(\bar{D}/8 + 2\bar{L} + \beta \bar{L}/16) + 2\beta^T L_g}{N} + 2S_T \xi + 2\beta^T \xi_T.$$

Choosing the spatial and numerical allocations as in the article proves the sufficient cap. It is not a statement that the fixed ladder 4, 8, 16 contains that cap, nor that the labels rounded to a fixed lattice support arbitrary tolerances. Acquisition, quantization and selector excess are additional summands in the same budget.

There are $(N + 1)^2$ nodes, $N + 1$ feasible actions per node, N shock midpoints per query and T backward dates. A direct cone query evaluates $(N + 1)^2$ pieces, while a bilinear query evaluates one local patch. The resulting operation bounds

are $O(TN^6)$ and $O(TN^4)$, respectively. These formulas include label formation, not only verification. Actor storage and critic-label storage are $O(TN^2)$. Circuit compilation, cell verification, selector queries, feasible repair, numerical enclosure, coefficient conversion and serialization are separately counted or bounded in the stated implementation. Arbitrary precision multiplies arithmetic counts by the cost of the chosen bit operations and requires enough guard precision to meet the query error. These are upper bounds for the specified routines, not lower bounds against alternative algorithms. In general dimension the original state and action covering factors remain visible. No computational saving follows from relabeling the identical envelope as a neural circuit.

14. DIRECT POLICY VALUES WITH NUMERICAL AND SAMPLING ERROR

14.1. Identity and deterministic support

For a fixed policy m and its own fitted continuation, let

$$\begin{aligned} Z_m = & f_0^m(X_0) + B_T(g(X_T) - f_T^m(X_T)) \\ & + \sum_{t < T} B_t \{c_t(X_t, A_t^m) + \beta_t \mathbb{E}[f_{t+1}^m(X_{t+1}) \mid \mathcal{H}_t] - f_t^m(X_t)\}. \end{aligned}$$

Conditional expectation and telescoping show $\mathbb{E}Z_m = J_m$ for either a fixed or random initial state. Policies in a pair may be driven by common shocks; preservation of each marginal law is sufficient. The innovation dependence between the two policies changes variance, not this identity.

For $f_t^m - \mathcal{T}_t f_{t+1}^m \leq u_t^m$ and $\mathcal{T}_t^{\pi_m} f_{t+1}^m - f_t^m \leq d_t^m$, the realized conditional residual is in $[-u_t^m, d_t^m]$. The lower bound follows because a feasible action cannot be below the Bellman infimum. Adding the ranges of these residuals, the terminal differences, and the initial difference gives the deterministic support in the article. For a common random initial state the initial difference is piecewise affine on the union of the two knot sets; extrema occur at those knots. For fixed initial states it is evaluated exactly as a rational number. Initial randomness is therefore not replaced by an unsupported average over four hand-selected points.

The frozen scalar terminal allowance is reconstructed from its stored width. If terminal node error is e_T , a cone continuation gives $g - f_T \in [-e_T - 2L_g h_T, e_T]$, and

a spline gives $g - f_T \in [-e_T - L_g h_T, e_T + L_g h_T]$. All signs are retained when forming pair support. Subtracting separate regret bounds would not give this estimand.

14.2. Bins, discontinuities, and endpoint statistics

Each independent uniform integer specifies one of 2^{40} equal innovation bins, and, for the uniform initial-state law, one such initial bin. The lower and upper path computation encloses every continuous point of that bin. Conditional uniform randomization within each independent bin recovers the original continuous uniform law. It need not be explicitly generated because every such value is already enclosed.

At a selector boundary, stored rational cuts are enclosed outward. Binary search obtains the first and last cells intersecting the state interval. The action interval is the hull of all actions in those cells, not merely its two endpoint actions. Propagating the hull may be conservative but is valid even for a nonmonotone actor. No path is removed. Intersections with the proved invariant state domain and deterministic score support preserve enclosure. The stored ambiguity counts quantify this part of the computation. These endpoint observations are independent across paths under the stated sampling model, even though endpoints and contrasts within a path are dependent.

Here are the numerical details of the empirical Bernstein calculation. For a binary64 endpoint sample $x_1, \dots, x_n \in [a, b]$, let $M = \max(|a|, |b|)$ and $\gamma = (n + 4)u / (1 - (n + 4)u)$. Denote the computed mean and mean square by $\tilde{\mu}, \tilde{q}$. The exact moments obey

$$\mu \in [\tilde{\mu} - \gamma M, \tilde{\mu} + \gamma M], \quad q \in [\tilde{q} - \gamma M^2, \tilde{q} + \gamma M^2].$$

The bound allows sequential summation, multiplication and division and is conservative for the implementation's pairwise sums. Rational arithmetic constructs these intervals from the exact binary64 outputs. There is no uncharged empirical-variance rounding. If $[\mu_-, \mu_+]$ is the mean enclosure, then

$$V^+ = \frac{n}{n-1} \left[q_+ - \min_{v \in [\mu_-, \mu_+]} v^2 \right]_+$$

is an upper bound for the unbiased sample variance. The minimum is zero if the mean interval contains zero and otherwise the smaller squared endpoint.

There are $K = 120$ contrasts, each with two one-sided tails, and total error $\alpha = 0.01$. The bound of [Maurer and Pontil \(2009\)](#) uses $\log(4K/\alpha) = \log(48000) < 11$. The inequality is verified by the exact positive sum $\sum_{j=0}^{40} 11^j/j! > 48000$. A valid endpoint radius is

$$R(V^+) = \sqrt{2V^+ 11/n} + \frac{7(b-a) 11}{3(n-1)}.$$

The code rounds the square-root result upward until its exact rational square is at least the radicand. All remaining additions are rational. Subtract this radius from the lower endpoint's mean lower bound and add it to the upper endpoint's mean upper bound. Intersect with known support, then round the final reported display outward. This proves simultaneous coverage with probability at least 0.99 conditional on the fixed-policy, independent-bin model. A seed supplies reproducibility, not evidence that a pseudorandom generator satisfies a mathematical independence axiom.

14.3. Decision margins and selection

The protocol fixed the superiority threshold at zero and the separate decision tolerance at $1/1024$ before the new comparison. Thus a full interval inside $[-1/1024, 1/1024]$ establishes two-sided cost proximity in the normalized units on the joint coverage event. It does not establish equality, a signed ranking, or a calibrated welfare equivalence. The theorem for choosing among a finite catalogue is a direct inequality on this same event: $J_{\hat{m}} - J_b \leq U_{\hat{m}}$ and $U_m \leq J_m - J_b + U_m - L_m$. Minimizing $U_m + k_m$ gives the stated oracle inequality. Charges are known inputs to that decision problem; converting observed seconds to an economic charge requires an additional model and is not performed here.

The complete cost intervals follow. The same fixed policy may appear under distinct target labels; its identity is retained in every row. A timing repetition is not a new trained policy. All lower and upper endpoint moments, supports, seeds,

stream hashes, ambiguity counts and input checkpoint hashes are deposited with the result records.

TABLE VI: All 120 direct expected-cost intervals

T	p	Target	Initial law	Contrast	Lower	Upper
2	1	1/4	1/2	C-S	-0.0002078	0.0001255
2	1	1/4	1/2	W-C	-0.0000865	0.0000825
2	1	1/4	1/2	W-S	-0.0001976	0.0001113
2	1	1/4	1/8	C-S	-0.0001577	0.0001711
2	1	1/4	1/8	W-C	-0.0000918	0.0000883
2	1	1/4	1/8	W-S	-0.0001472	0.0001572
2	1	1/4	7/8	C-S	-0.0001637	0.0001701
2	1	1/4	7/8	W-C	-0.0000785	0.0000718
2	1	1/4	7/8	W-S	-0.0001548	0.0001546
2	1	1/4	uniform	C-S	-0.0001858	0.0001574
2	1	1/4	uniform	W-C	-0.0000891	0.0001048
2	1	1/4	uniform	W-S	-0.0001656	0.0001529
2	1	1/8	1/2	C-S	-0.0000776	0.0000893
2	1	1/8	1/2	W-C	-0.0000443	0.0000393
2	1	1/8	1/2	W-S	-0.0000740	0.0000806
2	1	1/8	1/8	C-S	-0.0000908	0.0000736
2	1	1/8	1/8	W-C	-0.0000444	0.0000476
2	1	1/8	1/8	W-S	-0.0000830	0.0000690
2	1	1/8	7/8	C-S	-0.0000840	0.0000830
2	1	1/8	7/8	W-C	-0.0000387	0.0000365
2	1	1/8	7/8	W-S	-0.0000790	0.0000757
2	1	1/8	uniform	C-S	-0.0000850	0.0000870
2	1	1/8	uniform	W-C	-0.0000518	0.0000452
2	1	1/8	uniform	W-S	-0.0000821	0.0000776
2	1	1/16	1/2	C-S	-0.0000391	0.0000443

	T	p	Target	Initial law	Contrast	Lower	Upper	
1								1
2								2
3	2	1	1/16	1/2	W-C	-0.0000205	0.0000214	3
4	2	1	1/16	1/2	W-S	-0.0000356	0.0000417	4
5	2	1	1/16	1/8	C-S	-0.0000409	0.0000414	5
6	2	1	1/16	1/8	W-C	-0.0000230	0.0000226	6
7	2	1	1/16	1/8	W-S	-0.0000380	0.0000381	7
8	2	1	1/16	7/8	C-S	-0.0000459	0.0000377	8
9	2	1	1/16	7/8	W-C	-0.0000189	0.0000188	9
10	2	1	1/16	7/8	W-S	-0.0000429	0.0000346	10
11	2	1	1/16	uniform	C-S	-0.0000431	0.0000428	11
12	2	1	1/16	uniform	W-C	-0.0000280	0.0000207	12
13	2	1	1/16	uniform	W-S	-0.0000437	0.0000361	13
14	2	4	1/4	1/2	C-S	-0.0001589	0.0001743	14
15	2	4	1/4	1/2	W-C	-0.0000783	0.0000966	15
16	2	4	1/4	1/2	W-S	-0.0001367	0.0001704	16
17	2	4	1/4	1/8	C-S	-0.0001919	0.0001304	17
18	2	4	1/4	1/8	W-C	-0.0000923	0.0000953	18
19	2	4	1/4	1/8	W-S	-0.0001774	0.0001189	19
20	2	4	1/4	7/8	C-S	-0.0001889	0.0001489	20
21	2	4	1/4	7/8	W-C	-0.0000785	0.0000806	21
22	2	4	1/4	7/8	W-S	-0.0001750	0.0001370	22
23	2	4	1/4	uniform	C-S	-0.0001569	0.0001865	23
24	2	4	1/4	uniform	W-C	-0.0001162	0.0001049	24
25	2	4	1/4	uniform	W-S	-0.0001496	0.0001680	25
26	2	4	1/8	1/2	C-S	-0.0000783	0.0000884	26
27	2	4	1/8	1/2	W-C	-0.0000449	0.0000417	27
28	2	4	1/8	1/2	W-S	-0.0000735	0.0000804	28
29	2	4	1/8	1/8	C-S	-0.0000814	0.0000800	29
30	2	4	1/8	1/8	W-C	-0.0000458	0.0000491	30
31	2	4	1/8	1/8	W-S	-0.0000732	0.0000751	31
32	2	4	1/8	7/8	C-S	-0.0000899	0.0000793	32

	T	p	Target	Initial law	Contrast	Lower	Upper	
1								1
2								2
3	2	4	1/8	7/8	W-C	-0.0000385	0.0000408	3
4	2	4	1/8	7/8	W-S	-0.0000822	0.0000739	4
5	2	4	1/8	uniform	C-S	-0.0000792	0.0000925	5
6	2	4	1/8	uniform	W-C	-0.0000657	0.0000453	6
7	2	4	1/8	uniform	W-S	-0.0000829	0.0000758	7
8	2	4	1/16	1/2	C-S	-0.0000386	0.0000450	8
9	2	4	1/16	1/2	W-C	-0.0000215	0.0000222	9
10	2	4	1/16	1/2	W-S	-0.0000350	0.0000421	10
11	2	4	1/16	1/8	C-S	-0.0000380	0.0000427	11
12	2	4	1/16	1/8	W-C	-0.0000233	0.0000241	12
13	2	4	1/16	1/8	W-S	-0.0000343	0.0000398	13
14	2	4	1/16	7/8	C-S	-0.0000451	0.0000395	14
15	2	4	1/16	7/8	W-C	-0.0000200	0.0000196	15
16	2	4	1/16	7/8	W-S	-0.0000421	0.0000360	16
17	2	4	1/16	uniform	C-S	-0.0000404	0.0000457	17
18	2	4	1/16	uniform	W-C	-0.0000319	0.0000236	18
19	2	4	1/16	uniform	W-S	-0.0000413	0.0000384	19
20	4	1	1/4	1/2	C-S	-0.0001610	0.0001274	20
21	4	1	1/4	1/2	W-C	-0.0000776	0.0000826	21
22	4	1	1/4	1/2	W-S	-0.0001471	0.0001183	22
23	4	1	1/4	1/8	C-S	-0.0001324	0.0001547	23
24	4	1	1/4	1/8	W-C	-0.0000808	0.0000937	24
25	4	1	1/4	1/8	W-S	-0.0001144	0.0001497	25
26	4	1	1/4	7/8	C-S	-0.0001620	0.0001263	26
27	4	1	1/4	7/8	W-C	-0.0000784	0.0000726	27
28	4	1	1/4	7/8	W-S	-0.0001535	0.0001120	28
29	4	1	1/4	uniform	C-S	-0.0001608	0.0001330	29
30	4	1	1/4	uniform	W-C	-0.0000918	0.0000835	30
31	4	1	1/4	uniform	W-S	-0.0001535	0.0001175	31
32	4	1	1/8	1/2	C-S	-0.0000640	0.0000806	32

	T	p	Target	Initial law	Contrast	Lower	Upper	
1								1
2								2
3	4	1	1/8	1/2	W-C	-0.0000382	0.0000434	3
4	4	1	1/8	1/2	W-S	-0.0000556	0.0000774	4
5	4	1	1/8	1/8	C-S	-0.0000566	0.0000869	5
6	4	1	1/8	1/8	W-C	-0.0000395	0.0000476	6
7	4	1	1/8	1/8	W-S	-0.0000468	0.0000851	7
8	4	1	1/8	7/8	C-S	-0.0000724	0.0000719	8
9	4	1	1/8	7/8	W-C	-0.0000370	0.0000385	9
10	4	1	1/8	7/8	W-S	-0.0000660	0.0000669	10
11	4	1	1/8	uniform	C-S	-0.0000743	0.0000732	11
12	4	1	1/8	uniform	W-C	-0.0000441	0.0000435	12
13	4	1	1/8	uniform	W-S	-0.0000689	0.0000672	13
14	4	4	1/4	1/2	C-S	-0.0001615	0.0001292	14
15	4	4	1/4	1/2	W-C	-0.0000777	0.0000923	15
16	4	4	1/4	1/2	W-S	-0.0001419	0.0001241	16
17	4	4	1/4	1/8	C-S	-0.0001735	0.0001139	17
18	4	4	1/4	1/8	W-C	-0.0000906	0.0000909	18
19	4	4	1/4	1/8	W-S	-0.0001610	0.0001018	19
20	4	4	1/4	7/8	C-S	-0.0001368	0.0001565	20
21	4	4	1/4	7/8	W-C	-0.0000782	0.0000822	21
22	4	4	1/4	7/8	W-S	-0.0001225	0.0001462	22
23	4	4	1/4	uniform	C-S	-0.0001432	0.0001535	23
24	4	4	1/4	uniform	W-C	-0.0000963	0.0000992	24
25	4	4	1/4	uniform	W-S	-0.0001296	0.0001428	25
26	4	4	1/8	1/2	C-S	-0.0000647	0.0000808	26
27	4	4	1/8	1/2	W-C	-0.0000451	0.0000405	27
28	4	4	1/8	1/2	W-S	-0.0000609	0.0000724	28
29	4	4	1/8	1/8	C-S	-0.0000727	0.0000710	29
30	4	4	1/8	1/8	W-C	-0.0000444	0.0000464	30
31	4	4	1/8	1/8	W-S	-0.0000656	0.0000658	31
32	4	4	1/8	7/8	C-S	-0.0000731	0.0000734	32

T	p	Target	Initial law	Contrast	Lower	Upper
4	4	1/8	7/8	W-C	-0.0000405	0.0000394
4	4	1/8	7/8	W-S	-0.0000675	0.0000667
4	4	1/8	uniform	C-S	-0.0000700	0.0000788
4	4	1/8	uniform	W-C	-0.0000483	0.0000494
4	4	1/8	uniform	W-S	-0.0000634	0.0000732

W is the witness, C the same-critic nearest actor, and S the spline. Every displayed endpoint is rounded outward to seven decimal places; the exact rational intervals control all decisions and are deposited. All 120 intervals lie inside the predeclared band. Repeated target labels and policy hashes remain explicit in the underlying records.

15. COMPLETE FRONTIERS AND EXPERIMENT RECORDS

For each method, the first-crossing function is constant between adjacent values of its recorded certified bounds. Partitioning the positive tolerance axis by the union of both methods' bounds therefore gives the exact comparison everywhere, with left endpoints included because attainment is weak inequality. The full-prefix Bellman count charges every earlier failed rung. Above the largest breakpoint both methods use their first rung. An interval in which only one method attains is distinguished from an interval in which both attain with different query counts. This descriptive reconstruction is not a new experimental target design.

TABLE VII: Every interval of the scalar first-crossing curves

T	p	Tolerance lower	Tolerance upper	Relation	N_W/N_S
2	1	0.0000000	0.0343075	Neither	-/-
2	1	0.0343075	0.0396413	W only	512/-
2	1	0.0396413	0.0686151	Same	512/512

	T	p	Lower	Upper	Relation	N_W/N_S	
1							1
2							2
3	2	1	0.0686151	0.0791884	W fewer	256/512	3
4	2	1	0.0791884	0.1372302	Same	256/256	4
5	2	1	0.1372302	0.1579999	W fewer	128/256	5
6	2	1	0.1579999	0.2744603	Same	128/128	6
7	2	1	0.2744603	0.3144939	W fewer	64/128	7
8	2	1	0.3144939	0.5489206	Same	64/64	8
9	2	1	0.5489206	0.6229815	W fewer	32/64	9
10	2	1	0.6229815	1.0978413	Same	32/32	10
11	2	1	1.0978413	1.2220383	W fewer	16/32	11
12	2	4	0.0000000	0.0350171	Neither	–/–	12
13	2	4	0.0350171	0.0409569	W only	512/–	13
14	2	4	0.0409569	0.0700341	Same	512/512	14
15	2	4	0.0700341	0.0818309	W fewer	256/512	15
16	2	4	0.0818309	0.1400683	Same	256/256	16
17	2	4	0.1400683	0.1633408	W fewer	128/256	17
18	2	4	0.1633408	0.2801366	Same	128/128	18
19	2	4	0.2801366	0.3253087	W fewer	64/128	19
20	2	4	0.3253087	0.5602732	Same	64/64	20
21	2	4	0.5602732	0.6454998	W fewer	32/64	21
22	2	4	0.6454998	1.1205463	Same	32/32	22
23	2	4	1.1205463	1.2693757	W fewer	16/32	23
24	4	1	0.0000000	0.0643873	Neither	–/–	24
25	4	1	0.0643873	0.0663689	W only	512/–	25
26	4	1	0.0663689	0.1287747	Same	512/512	26
27	4	1	0.1287747	0.1325577	W fewer	256/512	27
28	4	1	0.1325577	0.2575494	Same	256/256	28
29	4	1	0.2575494	0.2643956	W fewer	128/256	29
30	4	1	0.2643956	0.5150987	Same	128/128	30
31	4	1	0.5150987	0.5259184	W fewer	64/128	31
32	4	1	0.5259184	1.0301974	Same	64/64	32

T	p	Lower	Upper	Relation	N_W/N_S
4	1	1.0301974	1.0403702	W fewer	32/64
4	1	1.0403702	2.0351529	Same	32/32
4	1	2.0351529	2.0603948	S fewer	32/16
4	4	0.0000000	0.0657205	Neither	–/–
4	4	0.0657205	0.0699794	W only	512/–
4	4	0.0699794	0.1314410	Same	512/512
4	4	0.1314410	0.1398014	W fewer	256/512
4	4	0.1398014	0.2628819	Same	256/256
4	4	0.2628819	0.2789813	W fewer	128/256
4	4	0.2789813	0.5257639	Same	128/128
4	4	0.5257639	0.5554218	W fewer	64/128
4	4	0.5554218	1.0515278	Same	64/64
4	4	1.0515278	1.1008786	W fewer	32/64
4	4	1.1008786	2.1030555	Same	32/32
4	4	2.1030555	2.1614686	W fewer	16/32

Lower endpoints are included and upper endpoints excluded in the exact records. Decimal breakpoint displays are descriptive rounded values, not substitute decision thresholds. The deposited rational endpoints determine the step functions without rounding ambiguity. Query counts equal the full prefix through the indicated rung; above each cell's last displayed upper endpoint both methods use their first rung.

TABLE VIII: All distinct constrained construction rungs

T	p	Method	N	Bound	Prefix queries	Midpoints
2	1	Bilinear FVI	4	10.12531	250	1,000
2	1	Bilinear FVI	8	5.37034	1,708	12,664
2	1	Bilinear FVI	16	2.79602	11,534	169,880
2	1	Witness	4	9.25915	250	1,000

T	p	Method	N	Bound	Prefix queries	Midpoints
2	1	Witness	8	4.62958	1,708	12,664
2	1	Witness	16	2.31479	11,534	169,880
2	4	Bilinear FVI	4	10.26707	250	1,000
2	4	Bilinear FVI	8	5.43656	1,708	12,664
2	4	Bilinear FVI	16	2.82885	11,534	169,880
2	4	Witness	4	9.47650	250	1,000
2	4	Witness	8	4.73825	1,708	12,664
2	4	Witness	16	2.36912	11,534	169,880
3	1	Bilinear FVI	4	14.09329	375	1,500
3	1	Bilinear FVI	8	7.45424	2,562	18,996
3	1	Bilinear FVI	16	3.89149	17,301	254,820
3	1	Witness	4	13.28483	375	1,500
3	1	Witness	8	6.64242	2,562	18,996
3	1	Witness	16	3.32121	17,301	254,820
3	4	Bilinear FVI	4	14.28140	375	1,500
3	4	Bilinear FVI	8	7.53684	2,562	18,996
3	4	Bilinear FVI	16	3.93210	17,301	254,820
3	4	Witness	4	13.64263	375	1,500
3	4	Witness	8	6.82132	2,562	18,996
3	4	Witness	16	3.41066	17,301	254,820

Each row has three isolated timing repetitions with identical policy and certificate checkpoint hashes. All 72 raw rungs remain deposited. Every state-dependent action net, repaired actor, continuation label and numerical allowance can be reconstructed from its checkpoint.

16. CONTINUOUS TENSOR COMPILATION AND FEASIBLE WITNESS IDENTITY

This section proves Theorem 14 of the article. We state its identities explicitly so that the numerical compiler can be checked without relying on an unproved

interpolation rule. The original labels and original witness indices, not transformed-node actions, are the inputs.

16.1. The line transform and its tie invariant

For ordered line locations $s_0 < \dots < s_m$, attach to each location a pair (q_j, a_j) , where q_j is a real cost and a_j an original index. Order such pairs lexicographically. Addition of a scalar to a pair adds it only to the cost. Start with $u_j = (q_j, a_j)$ and perform

$$\begin{aligned} u_j &\leftarrow \min_{\text{lex}} \{u_j, u_{j-1} + L(s_j - s_{j-1})\}, & j = 1, \dots, m, \\ u_j &\leftarrow \min_{\text{lex}} \{u_j, u_{j+1} + L(s_{j+1} - s_j)\}, & j = m - 1, \dots, 0. \end{aligned} \quad (1)$$

After the forward pass, induction gives $u_j = \min_{k \leq j, \text{lex}} (q_k + L(s_j - s_k), a_k)$. The backward pass makes u_j the minimum over all original line inputs of $(q_k + L|s_j - s_k|, a_k)$. To see this directly, a site to the right can be reached by the monotone backward path; a site to the left has already been considered in the forward pass. Any path that reverses direction has weakly larger distance because $L \geq 0$, so it cannot improve the cost. With $L = 0$ the original smallest index propagates in both directions. Lexicographic addition and minimum preserve the tie information throughout.

For a tensor grid, define the coordinate operator T_j by taking this minimum along coordinate j with all other coordinates fixed. Expanding the finite minima shows

$$T_d \cdots T_1(y_i, i)(v) = \min_{i, \text{lex}} (y_i + L\|v - x_i\|_1, i).$$

Thus the final result is (z_v, o_v) in (48). Each grid edge is relaxed at most twice in each coordinate. The edge count is at most dS , and all updates can be made in place with one cost and one index per node. The separable distance-transform part is the classical algorithm; see [Felzenszwalb and Huttenlocher \(2012\)](#). Original action owners are auxiliary information propagated by the same minimum operations, not independently recomputed choices.

16.2. *Off-grid extension and the original tie rule*

For any corner v and any original site i , the triangle inequality gives

$$y_i + L\|x - x_i\|_1 \leq y_i + L\|v - x_i\|_1 + L\|x - v\|_1.$$

Minimizing first over i proves the lower inequality in (49). For the converse, fix a site attaining $f(x)$. If its j th coordinate is at or to the left of the cell, choose the left corner coordinate; if it is at or to the right, choose the right corner coordinate. A grid site cannot lie strictly between adjacent grid coordinates. When x lies on the cell boundary, either admissible corner choice gives the same needed identity. Summing the coordinate identities gives

$$\|x - x_i\|_1 = \|x - v\|_1 + \|v - x_i\|_1.$$

Using this site in z_v proves the upper inequality and hence equality.

Now take the smallest attaining original index $i_*(x)$. For its associated corner v , equality in the preceding chain implies $z_v = y_{i_*} + L\|v - x_{i_*}\|_1$. If the original owner o_v were smaller than i_* , then

$$y_{o_v} + L\|x - x_{o_v}\|_1 \leq z_v + L\|x - v\|_1 = f(x).$$

It would be an attaining original site with a smaller index, a contradiction. Therefore $o_v = i_*(x)$. Evaluating the original cones of all corner owners and taking their lexicographic minimum proves (50). This argument covers inconsistent or dominated labels: it never uses $z_v = y_v$.

Attaching the original action a_{o_v} to a transformed label is essential. Attaching a_v need not preserve the policy. Under the established identity, both implementations select $i_*(x)$ and apply the same feasible repair $\Gamma_{i_*}(x)$. They consequently induce the same action at every state, the same controlled transition law and, by backward evaluation, the same policy value. The argument is not restricted to expected costs; identity of the actual policy preserves any well-defined common recursive evaluation. It does not equate two policies generated from different rounded label arrays.

16.3. Finite arithmetic and uncertain queries

All preprocessing inputs in the executable study are exact rational representations of the stored numerical labels, slopes and coordinate knots. Comparisons in (1) are exact. The compiler stores the resulting rational z_v and integer o_v . Its arithmetic-operation count does not suppress numerator/denominator growth: coefficient bit lengths and actual preprocessing time are recorded.

At an exact represented point, outward arithmetic encloses each term of (49); taking the minimum lower endpoint and the minimum upper endpoint encloses the true minimum. At an uncertain point $x \in D$, a midpoint value plus and minus $L \sup_{x \in D} \|x - \hat{x}\|_1$ is also a valid value enclosure. This is the continuation's Lipschitz property, not an actor regularity assertion.

For the actor, let $\mathcal{C}(D)$ be the finitely many tensor cells whose closures meet a query box. The previous proof implies that every exact winning original index at any point in that box belongs to $\{o_v : v \in \mathcal{V}(C), C \in \mathcal{C}(D)\}$. An interval lower bound above the smallest available upper bound rules out a candidate; all other candidates are retained. Their action hull includes the exact actor at every permitted state. Repair, downward action quantization, costs and transitions are then evaluated by interval extensions. This may enlarge the path enclosure, but cannot improve the statistical comparison by hiding a discontinuity. A degenerate ambiguous box is resolved by exact original-cone comparison with the same smallest-index tie rule.

The off-grid work is bounded by the corner count only for a point or one-cell query. A large uncertainty box can intersect many cells and requires their union. We do not use the point-query complexity theorem to claim dimension-independent or uncertainty-independent simulation work. The actual ambiguity and actor-query counts are included in the direct-comparison records.

16.4. Work to implemented economic accuracy

For Corollary 15, the deterministic primitive account gives $G \leq C_{T,p}/N + 2S_T\zeta + 2\beta^T\zeta_T$. Under its allocations the first term is at most $\varepsilon/4$ and the two numerical terms sum to at most $\varepsilon/4$. The exact-selector acquisition addition is $A2^{-b} + K\Delta \leq \varepsilon/2$, so the implemented loss is at most ε . A sufficient integer bit count can be found by

increasing b until the rational inequality holds; no unchecked logarithm is required. If observation has a fee, the augmented objective also pays $\lambda db S_T$. One cannot send b to infinity at fixed positive price and infer a vanishing augmented loss.

There are $(N + 1)^2$ state nodes, $N + 1$ actions at each node and N innovation midpoints. With a compiled continuation each midpoint requires a fixed number of two-dimensional primitive and corner operations, while transforming each date's labels and storing its feasible actors requires $O(N^2)$ further algebraic work. Thus the full backward construction has $O(TN^4)$ arithmetic operations. The dense alternative multiplies the midpoint workload by a further $(N + 1)^2$. The finite-bit actor applies the same compiled owner rule and a constant-dimensional capacity repair. For a dyadic ladder ending at N , $\sum_{j \geq 0} (N/2^j)^q \leq N^q / (1 - 2^{-q})$ for $q > 0$; this charges all previous construction attempts. The storage of the entire ladder of checkpoints also has the same $O(TN^2)$ scalar order.

These are sufficient algebraic upper bounds, not lower bounds for conventional methods. A p -bit elementary operation is charged at its actual or separately bounded arithmetic cost, and exact rational owners carry their coefficient sizes. Statistical policy comparison is outside this construction bound and is charged as its own joint service. No physical fsync latency, unknown optimizer iteration count, or unrestricted uncertainty-box enumeration is hidden inside the arithmetic order.

17. NONUNIFORM FITTED-VALUE COMPARISON AND MULTIDIMENSIONAL PRIMITIVES

17.1. A common all-state criterion on unequal coordinate grids

Let $g_t = \mathcal{T}_t f_{t+1}$ have graph modulus L_t . At nodes x_i , numerical feasible-net minimization returns y_i with $-e_t \leq y_i - g_t(x_i) \leq D_t^Q k_t + e_t$ and original action a_i with $Q_t(f_{t+1}; x_i, a_i) \leq y_i + e_t$. For multilinear interpolation on a rectangle of side lengths h_j^C , its nonnegative barycentric weights have sum one and satisfy

$$\sum_i w_i(x) \|x - x_i\|_1 = \sum_j \frac{2(x_j - x_j^-)(x_j^+ - x_j)}{h_j^C} \leq \frac{1}{2} \sum_j h_j^C \leq h_t,$$

where h_t is half the sum of the largest coordinate gaps. Hence the optimal-residual upper bound is $D_t^Q k_t + e_t + L_t h_t$.

Deploy the original action at the nearest coordinatewise node, choosing the smaller original index on a tie, with the same state-feasible repair. On its closed nearest-node cell D_i , feasible objective transport gives

$$Q_t(f_{t+1}; x, \Gamma_i(x)) - f_t(x) \leq e_t + y_i + L_t \|x - x_i\|_1 - f_t(x).$$

Intersect D_i with tensor cells and split at x_i in every coordinate. On each resulting rectangle, the expression after e_t is multi-affine: the absolute value has fixed sign and f_t is multilinear. A multi-affine function attains a maximum at a rectangle vertex, as repeated one-dimensional maximization shows. It therefore suffices to test the left midpoint, node, and right midpoint in each coordinate, at most 3^d vertices per node, using outward arithmetic. Denote this computed upper bound by E_t . The common one-sided policy criterion becomes

$$G_F = \sum_{t < T} B_t(D_t^Q k_t + 2e_t + L_t h_t + E_t) + B_T(2e_T + 2L_g h_T). \quad (2)$$

The witness criterion remains (41) with exact repair. Both compare against the original optimum over all admissible adapted policies. Neither treats a sampled residual as an all-state error.

The interpolant's ℓ^1 Lipschitz constant is the largest absolute edge slope: each partial derivative is a convex combination of edge slopes, and the dual norm of ℓ^1 is ℓ^∞ . Slopes are computed exactly from rational knots and labels. This modulus, not a neural modulus borrowed from another generator, is passed to that method's previous-date target construction.

17.2. The prespecified error-driven coordinate grid

The adaptive conventional comparator is a nonuniform tensor FVI method, not a sparse grid or locally adaptive quadtree. At every date and resolution $N > 4$, it computes a fresh 5^d pilot array using its own future continuation and the same N feasible action fractions and N innovation midpoints. At terminal date it evaluates the terminal primitive instead. These evaluations are charged even when a later grid repeats a pilot node.

For coordinate j , let $D_{j,k}$, $k = 1, 2, 3$, be the maximum, across all other pilot coordinates, of the absolute second difference $|y_{k-1} - 2y_k + y_{k+1}|$. Give coarse interval $[k/4, (k+1)/4]$, $k = 0, \dots, 3$, weight

$$w_{j,k} = 1 + 64 \max\{D_{j,\max(1,k)}, D_{j,\min(3,k+1)}\}.$$

Start with the four coarse intervals. Repeatedly bisect the interval maximizing $w_{j,k}$ times its squared length, breaking ties at the leftmost interval, until there are N intervals on that axis. All rankings and knots are rational. This positive curvature heuristic and its factor 64 were fixed in the scientific source before the full catalogue; no reported outcome selects them. The final cover uses the actual largest gaps. In particular, preferentially refining one area can worsen a global covering allowance elsewhere. The common certificate exposes that tradeoff rather than crediting the adaptive method with an assumed uniform radius.

17.3. Cycle-coupled dimensions two through four

The dimension stress study retains $x \in [0, 1]^d$, $a \in [0, \kappa(x)]$, with

$$\kappa(x) = \frac{1}{8} + \frac{1}{8d} \sum_i x_i, \quad F_i(x, a, z) = \frac{1}{16} + \frac{x_i}{2} + \frac{x_{i+1}}{8} + \frac{x_i(1 - x_{i+1})}{16} + b_i a + s_i z.$$

Indices are cyclic; b_i alternates $1/2, 1/4$, and s_i alternates $1, -1$. The shock is the original continuous uniform $z \in [-1/32, 1/32]$. Write $\bar{x} = d^{-1} \sum_i x_i$. The stage state cost is

$$\frac{2}{d} \sum_i (x_i - 5/8)^2 + \frac{1}{4d} \sum_i (x_i - x_{i+1})^2 + 2(1/2 - 2\bar{x})_+^2,$$

with action cost $pa^2 + 4a^4$. The terminal state cost doubles the first coefficient. At $d = 2$ these are exactly the original R47 primitives, including its action set and opposite innovation exposures.

For $d = 2, 3, 4$, valid ℓ^1 moduli are

$$C_x = \frac{13}{2d}, \quad L_g = \frac{9}{d}, \quad F_x = \frac{11}{16}, \quad F_a = \sum_i b_i, \quad C_a = \frac{p}{2} + \frac{1}{4}, \quad K^A = \frac{1}{8d}. \quad (3)$$

Indeed, $0 \leq a \leq 1/4$ and the displayed transition lies in $[1/32, 29/32]$ by direct bounds, so the unit cube is invariant. Each Jacobian column in the state has absolute sum at most $9/16 + 1/8 = 11/16$. The action and capacity moduli follow by differentiation.

For the state-cost bound, in the active shortage region the stage derivative with respect to x_j is

$$\frac{4}{d}(x_j - 5/8) + \frac{2x_j - x_{j-1} - x_{j+1}}{2d} - \frac{8}{d}(1/2 - 2\bar{x}).$$

At $d = 2$, the repeated neighbor is counted twice. Its coefficients in the coordinates are nonnegative for $d \leq 4$, so its minimum on the active region is attained at zero and equals $-13/(2d)$. Its positive part is at most the corresponding derivative without the negative shortage term, whose upper bound is $5/(2d)$. On the inactive region the derivative lies between $-7/(2d)$ and $5/(2d)$. The terminal derivative replaces $4/d$ by $8/d$; the analogous bounds give minimum $-9/d$ and smaller absolute upper bounds elsewhere. Continuity across the shortage boundary yields the global moduli in (3). These constants are proved for the displayed dimensions, not silently extrapolated to a different family.

A feasible action net has points $j\kappa(x)/N$, $j = 0, \dots, N$, and covering radius at most $1/(8N)$. In dimension three a downward binary64 realization adds at most 2^{-50} to that radius; its stored actions remain feasible. Dimensions two and four and the study's dyadic knots make these fractions exactly representable. Every path integral uses midpoint queries plus a continuous-law remainder. Since $\|F(x, a, z) - F(x, a, z')\|_1 = d|z - z'|$, averaging over M equal shock cells gives

$$\left| \mathbb{E}f(F(x, a, Z)) - \frac{1}{M} \sum_{m=1}^M f(F(x, a, z_m)) \right| \leq \frac{dL_f}{64M}.$$

The denominator is obtained from the mean absolute displacement within a cell, one quarter of its length. Discounting, interval evaluation and label rounding are additional terms; the deposited query error includes all of them.

18. OBSERVATION CHARGES AND DIRECT COST INTERVALS

18.1. *The sensor account and integer choice*

Each observed coordinate cell has width 2^{-b} and midpoint error at most $2^{-(b+1)}$, including the boundary cells. For the affine capacity, evaluating the lower endpoint of the observation box produces a feasible lower capacity. Its distance from the true capacity is at most $2K^A r(b)$. Downward action rounding by Δ adds at most Δ to repair displacement. Therefore the acquisition addition at date t is $2L_t r(b) + D_t^Q(2K_t^A r(b) + \Delta_t)$. Summing discounted additions and adding the deterministic observation charge gives (59). The same summation does not claim that a reference policy-loss bound is a policy-cost point estimate.

The integer precision comparisons follow from the successive difference $C - A2^{-(b+1)}$. If this difference vanishes, both b and $b + 1$ minimize locally, and monotonicity makes them global minimizers; the smaller is recorded. Every candidate precision, charge, allowance and augmented bound is stored as a rational number. Checking the two neighbors, or all candidates in the finite range, is an exact optimization certificate for this upper-bound objective.

18.2. *Continuous-law path enclosures*

For a uniform initial law, two independent forty-bit bins cover the initial coordinates. For a fixed initial vector, the coordinate inputs are exact. At each date a further independent bin encloses the common continuous innovation. The bin width is 2^{-40} for an initial coordinate and 2^{-44} for the shock interval of total length $1/16$.

For each controller separately, its state box determines all possible observation bins; exact midpoints of those bins form an observation enclosure. The exact actor at every possible observation is enclosed using original witness owners or the nearest-node FVI cells. Capacity repair uses the lowest capacity on each possible observation box. Its action hull includes all possible repaired and quantized actions. The natural interval extensions of cost and transition then propagate the enclosure to the next date. The compiler may accelerate these evaluations but never replaces a

path box by an unchecked midpoint actor. Dependence between the two policies is arbitrary inside a paired path; their marginals retain the prescribed innovation law.

The running-cost support follows from the separate bounds $\sum_j (x_j - 5/8)^2 \leq 25/32$, $(x_1 - x_2)^2/4 \leq 1/4$, $2(1/2 - x_1 - x_2)_+^2 \leq 1/2$, $pa^2 \leq p/16$, and $4a^4 \leq 1/64$. Terminal cost is at most $37/16$. These inequalities prove (62); separate maxima need not occur at the same state. The wider support is retained as a conservative input, rather than estimated from the simulated sample.

18.3. Moment and probability accounts

For a bounded endpoint array $X_1, \dots, X_n \in [a, b]$, write $M = \max(|a|, |b|)$, $u = 2^{-53}$ and $\gamma = (n + 4)u / (1 - (n + 4)u)$. The implementation encloses the computed mean by adding and subtracting γM ; it enlarges the computed mean square by γM^2 . These conservative bounds cover multiplication, summation and division in the finite arrays used here, with no overflow. If the mean enclosure is $[m_-, m_+]$ and the second-moment upper bound is q_+ , an unbiased sample-variance upper bound is

$$s_+^2 = \max \left\{ 0, \frac{n}{n-1} \left(q_+ - \min_{m \in [m_-, m_+]} m^2 \right) \right\}.$$

The minimum square is zero exactly when the interval contains zero. All moment bounds are converted to rationals before computing the confidence allowance. A square-root upper endpoint is checked by exact squaring and rounded upward until valid. The rational partial sum $\sum_{j=0}^{40} 10^j / j! > 19200 = 4K/\alpha$ verifies the logarithmic allowance $\ell = 10$.

The empirical-Bernstein inequality of [Maurer and Pontil \(2009\)](#), applied to the two bounded endpoint samples, gives (63)–(64). Each of the $2K$ one-sided assertions receives failure probability $\alpha / (2K)$. The union bound requires no independence across policy-pair tasks, but each task’s endpoint sample uses the stipulated independent-bin model. Stored PCG64 seeds and stream hashes give deterministic reproducibility only. The path enclosures are deterministic numerical statements; the confidence level is a separate sampling statement under that model.

For an installed FVI controller, changing to witness costs $k + D_{\mu,b}$. For an installed witness controller, changing to FVI costs $k - D_{\mu,b}$. The sign tests and no-recoupment decision in Theorem 20 follow. They concern the chosen transaction fee and initial law. They neither establish welfare equivalence under an unmodelled calibration nor prove that both controllers are close to the optimum merely because they are close to each other.

19. FULL WORK RECORDS AND PRESERVATION

For each completed service, let g_r be its exact bound at rung r and w_r its accumulated clock through that rung's durable checkpoint. The first-crossing work at tolerance ε is $w_{\min\{r:g_r \leq \varepsilon\}}$, with no attainment when this set is empty. Sorting the union of all exact bound breakpoints yields a complete partition of the positive tolerance axis into left-closed, right-open intervals on which every method's first crossing is fixed. Bounds need not be monotone for this construction: the earliest qualifying rung is always used. The lowest interval with no attainment and the unbounded upper interval are retained.

Every interval in the machine-readable frontier contains each method's selected resolution, median/minimum/maximum prefix wall time, CPU time, component counts and durable checkpoint size. The three repetitions are deterministic constructions with separate process clocks; they are not independent economic tasks or neural initializations. The dimension stress cells have one execution and are labelled accordingly. Peak resident memory includes the interpreter and warm-up. The internal prefix starts at primitives after the fixed non-economic warm-up; the parent process records the startup-inclusive duration separately. CPU affinity and numerical-library threads are fixed, while CPU frequency is not controlled.

The service includes target generation, adaptive pilots, exact compilation, continuous-law integration and its remainder, actor extraction, certification, failed rungs, serialization and a fixed finite acquired-deployment checksum. That checksum is a measured deployment component, not an empirical substitute for the uniform acquisition theorem. An independent direct policy-cost experiment is a separate joint service. Its clock includes both frozen-policy loads and compilation, hash checks, common paths, repair, statistics and durable output. This pair-specific

cost is reported separately; it is not assigned silently to one controller or called zero. Historical R46 scalar prefix-time curves are reconstructed from their original records without retiming or adding them to R48 clocks.

The scalar, original constrained and precision findings remain historical evidence. The new compiled construction is a separate prospective service, while the fixed-object compiler diagnostic retains the old labels and original witnesses. This separation prevents faster evaluation of a frozen object from being misreported as better optimizer reliability or a newly superior policy.

20. EVERY DIRECT CONSTRAINED-POLICY INTERVAL

Intervals concern witness minus FVI expected resource cost. The fee column records the two-direction no-recoupment decision at $k = 1/64$. The familywise statement is conditional on the independent-bin model. Displayed endpoints are rounded outward.

TABLE IX

DIRECT INTERVALS: HORIZON 2, PRICE 1

Initial law	Bits	Lower	Upper	Fee	Joint seconds
uniform	exact	-0.000821	0.001028	yes	2.77
uniform	6	-0.000738	0.001108	yes	2.71
uniform	10	-0.000806	0.001043	yes	2.72
1/8	exact	-0.001247	0.000596	yes	2.64
1/8	6	-0.001113	0.000726	yes	2.61
1/8	10	-0.001220	0.000622	yes	2.62
1/2	exact	-0.000935	0.000901	yes	2.62
1/2	6	-0.000918	0.000918	yes	2.59
1/2	10	-0.000933	0.000903	yes	2.62
7/8	exact	0.000869	0.002717	yes	2.64
7/8	6	0.000868	0.002716	yes	2.62
7/8	10	0.000869	0.002717	yes	2.64

Exact-state policies are unquantized. The finite-bit policies use robust capacity and downward action quantization. Joint seconds charge both policies, source loading, common simulation, interval statistics and durable output; they are not assigned to one method. The total-path numerical enclosure is distinct from sampling uncertainty.

TABLE X

DIRECT INTERVALS: HORIZON 2, PRICE 4

Initial law	Bits	Lower	Upper	Fee	Joint seconds
uniform	exact	-0.000694	0.001280	yes	2.76
uniform	6	-0.000687	0.001287	yes	2.73
uniform	10	-0.000694	0.001280	yes	2.73
1/8	exact	-0.001142	0.000830	yes	2.62
1/8	6	-0.001014	0.000959	yes	2.63
1/8	10	-0.001151	0.000821	yes	2.62
1/2	exact	-0.000136	0.001840	yes	2.63
1/2	6	-0.000374	0.001602	yes	2.65
1/2	10	-0.000145	0.001832	yes	2.61
7/8	exact	-0.000982	0.000983	yes	2.62
7/8	6	-0.000982	0.000982	yes	2.62
7/8	10	-0.000982	0.000983	yes	2.64

Exact-state policies are unquantized. The finite-bit policies use robust capacity and downward action quantization. Joint seconds charge both policies, source loading, common simulation, interval statistics and durable output; they are not assigned to one method. The total-path numerical enclosure is distinct from sampling uncertainty.

21. COMPLETE FIRST-CERTIFICATE TIME PARTITIONS

Each row is a left-closed, right-open tolerance interval. At the exact rational left endpoint, each method uses its first qualifying checkpoint; a dash denotes nonattainment. The final interval extends to infinity. Displayed endpoints have six significant digits; exact boundaries and complete component counts are in the deposited machine-readable frontier. Times are median prefix seconds for two-state services and singleton prefix seconds for dimension stress cases. These partitions are retrospective descriptions of the frozen complete ladders, not new prospective targets.

REFERENCES

FELZENSZWALB, PEDRO F. AND DANIEL P. HUTTENLOCHER (2012): "Distance Transforms of Sampled Functions," *Theory of Computing*, 8 (19), 415–428. []

TABLE XI

DIRECT INTERVALS: HORIZON 3, PRICE 1

Initial law	Bits	Lower	Upper	Fee	Joint seconds
uniform	exact	-0.001028	0.001281	yes	3.95
uniform	6	-0.000912	0.001394	yes	3.87
uniform	10	-0.001007	0.001302	yes	3.88
1/8	exact	-0.001501	0.000801	yes	3.79
1/8	6	-0.001364	0.000935	yes	3.76
1/8	10	-0.001455	0.000846	yes	3.73
1/2	exact	-0.001217	0.001079	yes	3.77
1/2	6	-0.001147	0.001147	yes	3.76
1/2	10	-0.001216	0.001080	yes	3.76
7/8	exact	0.000322	0.002627	yes	3.78
7/8	6	0.000223	0.002528	yes	3.78
7/8	10	0.000334	0.002639	yes	3.79

Exact-state policies are unquantized. The finite-bit policies use robust capacity and downward action quantization. Joint seconds charge both policies, source loading, common simulation, interval statistics and durable output; they are not assigned to one method. The total-path numerical enclosure is distinct from sampling uncertainty.

MAURER, ANDREAS AND MASSIMILIANO PONTIL (2009): "Empirical Bernstein Bounds and Sample Variance Penalization," in *Proceedings of the 22nd Annual Conference on Learning Theory*, arXiv:0907.3740. []

TABLE XII

DIRECT INTERVALS: HORIZON 3, PRICE 4

Initial law	Bits	Lower	Upper	Fee	Joint seconds
uniform	exact	-0.000688	0.001806	yes	3.92
uniform	6	-0.000709	0.001783	yes	3.90
uniform	10	-0.000685	0.001808	yes	3.90
1/8	exact	-0.001356	0.001135	yes	3.78
1/8	6	-0.001342	0.001149	yes	3.78
1/8	10	-0.001365	0.001126	yes	3.77
1/2	exact	-0.001114	0.001373	yes	3.80
1/2	6	-0.001048	0.001440	yes	3.78
1/2	10	-0.001111	0.001376	yes	3.77
7/8	exact	-0.000916	0.001570	yes	3.83
7/8	6	-0.000918	0.001568	yes	3.80
7/8	10	-0.000916	0.001569	yes	3.79

Exact-state policies are unquantized. The finite-bit policies use robust capacity and downward action quantization. Joint seconds charge both policies, source loading, common simulation, interval statistics and durable output; they are not assigned to one method. The total-path numerical enclosure is distinct from sampling uncertainty.

TABLE XIII

FULL TOLERANCE PARTITION: DIMENSION 2, HORIZON 2, PRICE 1

ε_L	ε_R	N_C	C (s)	N_U	U (s)	N_A	A (s)
0	0.578697	–	–	–	–	–	–
0.578697	0.718896	64	47.996	–	–	–	–
0.718896	1.15739	64	47.996	64	36.303	64	36.684
1.15739	1.42452	32	3.686	64	36.303	64	36.684
1.42452	2.31479	32	3.686	32	2.924	32	3.052
2.31479	2.79602	16	0.346	32	2.924	32	3.052
2.79602	4.62958	16	0.346	16	0.318	16	0.363
4.62958	5.37034	8	0.055	16	0.318	16	0.363
5.37034	9.25915	8	0.055	8	0.077	8	0.092
9.25915	10.1253	4	0.015	8	0.077	8	0.092
10.1253	∞	4	0.015	4	0.030	4	0.030

The all-state loss certificate is the common stopping criterion. Prefix time includes all earlier failed rungs, own-future construction, integration, compilation or adaptive pilots, repair checks and check-point fsync. Independent policy-cost inference is a separate joint service. The adaptive-coordinate comparator is executed only in dimension two.

TABLE XIV

FULL TOLERANCE PARTITION: DIMENSION 2, HORIZON 2, PRICE 4

ε_L	ε_R	N_C	C (s)	N_U	U (s)	N_A	A (s)
0	0.592281	–	–	–	–	–	–
0.592281	0.727139	64	48.070	–	–	–	–
0.727139	1.18456	64	48.070	64	36.386	64	37.020
1.18456	1.44098	32	3.671	64	36.386	64	37.020
1.44098	2.36912	32	3.671	32	2.936	32	3.063
2.36912	2.82885	16	0.344	32	2.936	32	3.063
2.82885	4.73825	16	0.344	16	0.318	16	0.363
4.73825	5.43656	8	0.055	16	0.318	16	0.363
5.43656	9.4765	8	0.055	8	0.078	8	0.093
9.4765	10.2671	4	0.015	8	0.078	8	0.093
10.2671	∞	4	0.015	4	0.030	4	0.030

The all-state loss certificate is the common stopping criterion. Prefix time includes all earlier failed rungs, own-future construction, integration, compilation or adaptive pilots, repair checks and check-point fsync. Independent policy-cost inference is a separate joint service. The adaptive-coordinate comparator is executed only in dimension two.

TABLE XV

FULL TOLERANCE PARTITION: DIMENSION 2, HORIZON 3, PRICE 1

ε_L	ε_R	N_C	C (s)	N_U	U (s)	N_A	A (s)
0	0.830302	–	–	–	–	–	–
0.830302	1.00278	64	71.928	–	–	–	–
1.00278	1.6606	64	71.928	64	54.643	64	55.672
1.6606	1.98559	32	5.534	64	54.643	64	55.672
1.98559	3.32121	32	5.534	32	4.366	32	4.560
3.32121	3.89149	16	0.503	32	4.366	32	4.560
3.89149	6.64242	16	0.503	16	0.473	16	0.533
6.64242	7.45424	8	0.078	16	0.473	16	0.533
7.45424	13.2848	8	0.078	8	0.111	8	0.133
13.2848	14.0933	4	0.021	8	0.111	8	0.133
14.0933	∞	4	0.021	4	0.043	4	0.043

The all-state loss certificate is the common stopping criterion. Prefix time includes all earlier failed rungs, own-future construction, integration, compilation or adaptive pilots, repair checks and check-point fsync. Independent policy-cost inference is a separate joint service. The adaptive-coordinate comparator is executed only in dimension two.

TABLE XVI

FULL TOLERANCE PARTITION: DIMENSION 2, HORIZON 3, PRICE 4

ε_L	ε_R	N_C	C (s)	N_U	U (s)	N_A	A (s)
0	0.852664	–	–	–	–	–	–
0.852664	1.01307	64	71.958	–	–	–	–
1.01307	1.70533	64	71.958	64	54.146	64	55.021
1.70533	2.00604	32	5.498	64	54.146	64	55.021
2.00604	3.41066	32	5.498	32	4.367	32	4.550
3.41066	3.9321	16	0.499	32	4.367	32	4.550
3.9321	6.82132	16	0.499	16	0.472	16	0.530
6.82132	7.53684	8	0.077	16	0.472	16	0.530
7.53684	13.6426	8	0.077	8	0.111	8	0.133
13.6426	14.2814	4	0.021	8	0.111	8	0.133
14.2814	∞	4	0.021	4	0.043	4	0.043

The all-state loss certificate is the common stopping criterion. Prefix time includes all earlier failed rungs, own-future construction, integration, compilation or adaptive pilots, repair checks and check-point fsync. Independent policy-cost inference is a separate joint service. The adaptive-coordinate comparator is executed only in dimension two.

TABLE XVII

FULL TOLERANCE PARTITION: DIMENSION 3, HORIZON 2, PRICE 1

ε_L	ε_R	N_C	C (s)	N_U	U (s)	N_A	A (s)
0	2.32986	–	–	–	–	–	–
2.32986	2.81383	16	9.499	–	–	–	–
2.81383	4.65973	16	9.499	16	6.671	–	–
4.65973	5.43228	8	0.551	16	6.671	–	–
5.43228	9.31945	8	0.551	8	0.539	–	–
9.31945	10.3112	4	0.051	8	0.539	–	–
10.3112	∞	4	0.051	4	0.107	–	–

The all-state loss certificate is the common stopping criterion. Prefix time includes all earlier failed rungs, own-future construction, integration, compilation or adaptive pilots, repair checks and check-point fsync. Independent policy-cost inference is a separate joint service. The adaptive-coordinate comparator is executed only in dimension two.

TABLE XVIII

FULL TOLERANCE PARTITION: DIMENSION 3, HORIZON 3, PRICE 1

ε_L	ε_R	N_C	C (s)	N_U	U (s)	N_A	A (s)
0	3.34922	–	–	–	–	–	–
3.34922	3.91479	16	13.900	–	–	–	–
3.91479	6.69844	16	13.900	16	9.900	–	–
6.69844	7.54254	8	0.801	16	9.900	–	–
7.54254	13.3969	8	0.801	8	0.799	–	–
13.3969	14.3472	4	0.072	8	0.799	–	–
14.3472	∞	4	0.072	4	0.157	–	–

The all-state loss certificate is the common stopping criterion. Prefix time includes all earlier failed rungs, own-future construction, integration, compilation or adaptive pilots, repair checks and check-point fsync. Independent policy-cost inference is a separate joint service. The adaptive-coordinate comparator is executed only in dimension two.

TABLE XIX

FULL TOLERANCE PARTITION: DIMENSION 4, HORIZON 2, PRICE 1

ε_L	ε_R	N_C	C (s)	N_U	U (s)	N_A	A (s)
0	4.62958	–	–	–	–	–	–
4.62958	5.45577	8	7.728	–	–	–	–
5.45577	9.25915	8	7.728	8	7.362	–	–
9.25915	10.3892	4	0.382	8	7.362	–	–
10.3892	18.2484	4	0.382	4	0.945	–	–
18.2484	18.5183	4	0.382	2	0.307	–	–
18.5183	∞	2	0.043	2	0.307	–	–

The all-state loss certificate is the common stopping criterion. Prefix time includes all earlier failed rungs, own-future construction, integration, compilation or adaptive pilots, repair checks and check-point fsync. Independent policy-cost inference is a separate joint service. The adaptive-coordinate comparator is executed only in dimension two.

TABLE XX

FULL TOLERANCE PARTITION: DIMENSION 4, HORIZON 3, PRICE 1

ε_L	ε_R	N_C	C (s)	N_U	U (s)	N_A	A (s)
0	6.64242	–	–	–	–	–	–
6.64242	7.58853	8	11.544	–	–	–	–
7.58853	13.2848	8	11.544	8	10.976	–	–
13.2848	14.4743	4	0.549	8	10.976	–	–
14.4743	25.2494	4	0.549	4	1.387	–	–
25.2494	26.5697	4	0.549	2	0.454	–	–
26.5697	∞	2	0.061	2	0.454	–	–

The all-state loss certificate is the common stopping criterion. Prefix time includes all earlier failed rungs, own-future construction, integration, compilation or adaptive pilots, repair checks and checkpoint fsync. Independent policy-cost inference is a separate joint service. The adaptive-coordinate comparator is executed only in dimension two.

TABLE XXI

HISTORICAL SCALAR FRONTIER USING ITS ORIGINAL COMPLETE-PREFIX CLOCKS

T	p	W earlier	S earlier	Tie	W only	S only	Neither
2	1	5	12	0	1	0	1
2	4	5	12	0	1	0	1
4	1	4	13	0	1	0	1
4	4	5	12	0	1	0	1

Counts are tolerance intervals, not probabilities or weighted economic tasks. W is the R46 witness and S its spline comparator. These are unaltered historical internal prefix clocks, not R48 timings; they are never added to new compiler or direct-comparison clocks. The exact partitions, selected checkpoints and all three repeated times are retained.